

claude-windows / ac48091f-a4f4-4268-b968-75205d9a50ff

Metadata

```
- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\ac48091f-a4f4-4268-b968-75205d9a50ff\subagents\workflows\wf_6abc9156-811\agent-ad99af7524871ad9f.jsonl`
- SHA-256: `c6082adcc0394c87ebc387c6f6ef184685aeba8deb2551452f55b0836db9e9d0`
- Source modified: `2026-06-22T02:04:01+00:00`
- Imported at: `2026-07-05T16:48:24+00:00`
- project: `wf_6abc9156-811`
- session_id: `ac48091f-a4f4-4268-b968-75205d9a50ff`
```

Transcript

1. user (2026-06-22T01:55:49.373Z)

You are scanning planning docs to surface ACCURATE, prioritized next steps for a co-founder.

REPO RULES:

```
- Engine repo at /c/Users/avidu/Projects/badminton-highlight-indexer: its working tree is on a
STALE serving branch. Read DOCS + verify SHIPPED state from ENGINE origin/master ONLY:
  git -C /c/Users/avidu/Projects/badminton-highlight-indexer show origin/master:docs/FILE.md
(read a doc on master; if it errors, the doc may be branch-local ? fall back to the working-tree
file and SAY SO)
  git -C /c/Users/avidu/Projects/badminton-highlight-indexer log origin/master --oneline -60
(what actually shipped ? use this to catch stale "TODO/next" claims)
  git -C /c/Users/avidu/Projects/badminton-highlight-indexer grep -n "PATTERN" origin/master
-- 'PATHGLOB'
- khelsutra repo at /c/Users/avidu/Projects/khelsutra: on clean master ? use Read/Grep/Glob
directly + git -C /c/Users/avidu/Projects/khelsutra log --oneline -40.
```

ANTI-SPIRAL (critical): before listing any "[gap]/net-new/TODO/next" item, VERIFY it hasn't already shipped ? cross-check the doc's claim against git log origin/master. A doc saying "X is next" is a HYPOTHESIS until you confirm X isn't already merged. Flag any stale doc claims you correct.

Context (already TRUE as of this session, do not re-list as "next"): the khelsutra ALPHA is DEPLOYED + live on a CPU droplet (engine + cloudflared tunnel + Cloudflare Access + SPA on Pages, real Gemini key working, full chain verified). DEPLOY_ALPHA fixes + the droplet kit are merged (khelsutra #78/#79/#81/#85). T2 (alpha deploy) is effectively DONE.

Return prioritized next items with honest status + git grounding.

TRACK: ****this-session follow-ups + cross-cutting hygiene****. Consider what THIS session leaves open: (a) ****Finding B**** ? the engine returns job=done with 0 rallies + the missing-Gemini-key error only in the log, not the job status (should it fail the job?) ? is there an existing tracking doc/issue, and where does it belong (serving owns backend/config + the gemini segmenter)? (b) the ****alpha-deploy follow-ups****: native-speaker review before WIDE launch, retention sweep scheduling (engine #301), monitoring/observability of the live droplet. Read ENGINE origin/master `docs/DOC\_STATUS.md` + `docs/BACKLOG.md` (if present) + `docs/NEXT\_STEPS.md`. Also note the broad CF token the owner should revoke.
ANSWER: prioritized follow-ups + any doc-hygiene/observability items. Ground in git/docs.

2. assistant (2026-06-22T01:55:52.884Z)

I'll scan the planning docs and verify shipped state. Let me start by reading the key docs from engine origin/master and checking git history.

3. user (2026-06-22T01:55:58.077Z)

=== ENGINE LOG origin/master ===

a42c132 docs(pkg): track the batteries-included app plan (PACKAGING_PLAN.md) + index pointers (#309)
de2bf14 feat(pkg): P2e ? lazy-torch so the LIGHT tier imports with NO torch installed (default-OFF) (#308)
679e642 feat(pkg): P2d ? DB upgrade-safety (downgrade guard + backup-before-migrate + export) (default-OFF) (#307)
f01dc85 feat(db): v7 ? record the UI locale per job (PMF analytics) + count_jobs_by_locale (#306)
fa7ee3f feat(pkg): P2a ? single ffmpeg/ffprobe resolver (env-overridable; one fail-loud place) (default-OFF) (#305)
e99d0c3 feat(stitch): localize the reel watermark to the UI locale (+ OFL Noto fonts) (#304)
ca34787 feat(pkg): P2b ? friendly `indexer --app` launcher (boot localhost server + open the bundled UI) (#303)
4a06b8b feat(pkg): P1b ? bundle the KhelSutra SPA single-origin into the engine wheel (default-OFF) (#302)
be0c434 feat(pkg): P1a ? net-new console/wizard endpoints: /api/version, /api/capabilities, /api/videos, /api/reels (#300)
cb7394b feat(serving): T11 ? retention sweep for stale uploads + caches (#299)
c039b90 feat(pkg): P0e ? single-instance lock so a 2nd process can't corrupt in-flight job state (default-OFF) (#298)
e8bad40 feat(pkg): P0d ? declare scipy (undeclared serving dep) + license-scan & built-wheel CI lanes (#297)
3895963 feat(pkg): P0c ? config/profile-driven bind address + DNS-rebinding Host-header guard (default-OFF) (#296)
fc78fce feat(pkg): P0b ? redact secret-shaped fields from the unauthenticated GET /api/config (#295)
0c0e064 feat(serving): exposure-safety boot posture ? fail fast on a half-configured exposed engine (default-OFF) (#294)
82a28f4 feat(pkg): P0a ? resolved app-data HOME (RALLY_APP_HOME) for config/.env/DB/output (default-OFF) (#293)
2afbadf fix(dryrun): install the watermark font in bootstrap + track the UI-trigger asks (#292)
7835f34 feat(gcp): always-install Ops Agent on GPU VMs + a cloud-GPU dry-run guide (#291)
aac7633 feat(serving): M9-auth ? Cloudflare Access edge-auth + per-tenant owner_id (default-OFF) (#290)
e68b583 fix(api): resolve a bucket-ingested video's source to a local file before stitching (#289)
43eccc1 feat(serving): M8 ? per-job cost ledger + versioned pricebook + /cost API (default-OFF) (#288)
8cdf65c feat(serving): M6 ? Cloud Run GPU dispatch shim + worker image (default-OFF, mocked) (#287)
b5af66e feat(serving): M5 ? truly-local profile per-profile startup checks + L4 profile-parity test (default-OFF) (#286)
fefa412 docs(serving): resolve §8 design picks (D16/D17) + reconcile per-video P3 vs M3 (#285)
06ff4b7 test(api): lock gs://|gcs:// /api/process -> queryable API-DB video_id (B-P1 guard) (#284)
cee4baa feat(serving): M4 ? DeviceContext + native WASB CPU-fallback (device="auto", cuda stays strict) (#283)
2afd54f feat(serving): M3 ? configurable output/scratch base (de-hardcode output/, default-OFF) (#282)
41d45fd fix(server): drain async jobs under `python -m backend.main` (double-module hang) (#281)
e111add feat(serving): M2 ? URI-aware input (input_backend/input_root, local=identity) (#280)
ce22791 feat(serving): M1 ? deployment.profile + compute_target config (pure, additive, default-OFF) (#279)
f518e97 docs(serving): M0 APPROVED + lock D13-D15 (switchable profiles, Drive Northstar, consent-not-surprise) (#278)
49c082a docs(serving): lock the 5 M0 gating decisions (D7-D12) + add the data-flywheel harness (M11) (#277)
f592281 docs: point I18N_PLAN + UI_PROPOSAL (+ DOC_STATUS) at the khelsutra UX/l10n project (#275)
f6f15ca docs: isolate field-ops/PMF + Roora into archives/; carve a no-ML docs/data_pipeline/ (#276)
03fa951 docs: resolve DOC_STATUS findings #1 (per-video supersede) + #2 (NEXT_STEPS refresh) (#274)
aa5e9db docs: add DOC_STATUS register + archival due-diligence convention (#273)

bd42250 docs: spin up COMPUTE_DECOUPLED_SERVING subproject + ADR-014 (#272)
4957b72 fix(worker): configure logging + explain a 0-segment infer (the cold-start blind spot) (#271)
4134721 docs: archive DECOUPLED_COMPUTE ? DELIVERED (M0-M2 + M3.5 on GCP) (#270)
c8e60ee fix(gen0): provenance-stamp the manifest git_sha off a git-archive tarball (#269)

4. user (2026-06-22T01:55:59.779Z)

=== KHELSUTRA LOG ===
17242a9 feat(site): localize own-model body content to kn/hi (#86)
7b04f02 deploy(kit): align with the live-verified droplet deploy (+ the gemini-default fix) (#85)
5e5a2ac docs(friendly-multilingual): close out the chapter (\$9 DoD + UI/UX + B7 deferred) (#84)
4c0f35c fix(site): UI/UX ? compact language switcher + header overflow fix + localized hero (#83)
0fae364 feat(web+site): A2 ? localized WhatsApp share for the reel (SPA + site) (#82)
1e52cfd deploy(droplet): tailored alpha kit for the CPU droplet (engine api. side) (#81)
64076a5 test(web): add a coverage floor (CI-gated) + i18n regression tests for the #77 audit (#80)
8328721 docs(tracker): refresh the \$7 milestone prose to merged reality (A6/A7/#77 landed) (#79)
3f0cdb8 docs(deploy): fix two would-break-the-deploy gaps in the alpha runbook (RALLY_ALLOWED_HOSTS + gs:// jail) (#78)
9f1ec9c fix(i18n): native-review pass on the kn/hi SPA console catalogs (#77)
9a2d7d2 feat(site): A7 ? camera-placement diagram on the capture page (#76)
aa8d5e9 feat(site): A6 ? plain-language Request-Access GPU/storage questions (#75)
1cd84af feat(site): A3 ? localize recording-education + marketing to kn/hi, Noto on the site (#74)
c1848df feat(web): LW2 ? send the UI locale to /api/compile (watermark) + /api/process (analytics) (#73)
1b1fa7b feat(web+site): B3 ? trilingual marketing site via shared t()/data-i18n shim (D-CATALOG-HOME) (#70)
f7e2bd5 chore(site): add robots.txt + a "crawling / bot protection" runbook for khelsutra.guru (#71)
bea8a2a docs(appliance): flip T11 retention sweep to ? ? tool shipped (engine #299), scheduling tracked (#301) (#69)
46e0244 feat(web): A9 ? localized tap-only filter controls (#68)
a2292ef feat(web): MULTILINGUAL Phase A ? Spanish/Danish/Indonesian SPA catalogs (#67)
ca619fb feat(web): T4 ? in-UI file upload (the runnable-tool keystone) (#66)
4627830 docs(deploy): alpha deploy kit ? make Studio a shareable page (split: Pages + api.* tunnel + CF Access) (#65)
8bbc557 feat(web): VITE_API_BASE ? let the SPA target a split api.* origin (default same-origin /api) (#64)
92a6c45 feat(web): A8 ? background-processing reassurance + icon+text status chips (#63)
501c014 feat(web): A5 ? friendly engine-error mapping across build + load (shared errors.*) (#62)
95195a5 feat(web): A4 ? mobile touch targets (?44px) + visible tooltip text (#61)
ccdcfd feat(web): A1 ? plain-language copy pass on the en catalog (de-jargon) (#60)
a20002f docs: propose Multilingual Expansion ? 7 new locales (th/id/da/es/zh-Hans/ja/ko) (#58)
a0475bc docs: hygiene refresh ? sync trackers to shipped reality (#59)
7750afc docs(dry-run): Path-B live e2e DONE ? in-UI bucket?reel CUJ + screencast (DoD \$6 box 3 ?) (#57)
4525c5d feat(web): B-P2 ? in-UI bucket ingest (point at a gs:// video ? process ? load) (#56)
d0461fd docs: correct stale "bucket ingest is net-new" claims (B-P1 already shipped in M2/#280) (#55)
2421557 docs(dry-run): record Path-A local result (en+hi reel downloaded) + the engine fix it surfaced (#54)
113587f docs: set the dry-run CUJ as a tracked goal (UI drives a bucket video -> reel download, multilingual) (#53)
1b69e1a feat(site): show a localized UI screenshot on each academy flyer (#52)
3a42847 feat(web): localized sample-query prompt (en/kn/hi) + i18n-test rule for query parsing (#51)
0021c4f test(web): guard the SPA stays self-contained / offline-renderable (#50)
631c3aa docs(friendly-multilingual): SPA fully trilingual ? mark B1/B2/B4/B5/B6 done + screencast milestone (#49)
ad0d7ed feat(web): ship the Hindi (hi) catalog ? tracker B6 (the extensibility proof) (#48)

4449981 feat(web): ship the Kannada (kn) catalog ? tracker B5 (#47)
a423b82 feat(web): self-host OFL Noto Kannada + Devanagari fonts ? tracker B4 (#46)

5. assistant (2026-06-22T01:56:03.628Z)

Good grounding so far. The engine M9-auth, T11 retention, exposure-safety, and P0-P2 packaging all shipped. Now let me read the key docs: DOC_STATUS.md, BACKLOG.md, NEXT_STEPS.md from engine origin/master.

6. user (2026-06-22T01:56:07.143Z)

```
=== DOC_STATUS exists? ===  
YES  
=== BACKLOG exists? ===  
YES  
=== NEXT_STEPS exists? ===  
YES
```

7. user (2026-06-22T01:56:11.191Z)

Next steps (live) ? refreshed 2026-06-21

> **How to read:** this top section is the **current cross-track state + prioritized pending list** (refreshed 2026-06-21, replacing the stale 2026-06-13 header). The detailed **owned-model & rally-quality track** plan (2026-06-13) is **preserved below, still valid** ? it is now **one of several* active tracks**. Live blow-by-blow: memory ``current-state.md``. Doc lifecycle/hygiene: ``docs/DOC_STATUS.md``.

? Where we are (2026-06-21)

DECOUPLED_COMPUTE delivered on GCP + archived* (#270); **compute = GCP only** (ADR-013, DO dropped). **COMPUTE_DECOUPLED_SERVING`** subproject spun up (#272, **ADR-014**) ? one pure core + local/cloud profiles; **owner M0 design sign-off pending**. **Per-video workspace P0** shipped (#264); **doc-status register + archival due-diligence** in place (#273). The **khelsutra site + per-rally UI** ship in parallel (sibling track). The **rally-quality / owned-model** track (below) is the core ML work, gated on **growing the golden corpus**.

? Prioritized pending ? all tracks

P0 ? highest leverage / unblocks the most

0. **[owner-priority, 2026-06-21] MAKE IT UI-DRIVEN + ALPHA-SHAREABLE** ? "a few recordings lying around ? highlights, from a page, no SSH". Today the cloud flow is **SSH/CLI-only** (``docs/CLOUD_GPU_DRYRUN_GUIDE.md``); **M12 does NOT fix it**. Needs (a separate khelsutra ``web/`` SPA + deploy track, NOT an engine seam): **(i)** deploy the engine as a reachable **service** (M7 ? Cloud Run service, ? owner GCP spend for a **persistent* endpoint**), **(ii)** the **SPA ingest/progress screen** (khelsutra ``web/`` B-P2: enter/upload video ? ``/api/process`` ? poll ``/api/jobs`` ? reel; ZERO engine code), **(iii)** flip **M9 edge-auth ON** (merged #290, default-OFF) to gate alpha tenants, **(iv)** a **compute PICKER** in the UI ("your machine vs our cloud", D13/D15). The engine seams make it **possible**; the SPA makes it **usable**. Full record: `[[ui-driven-cloud-serving-gap]]`.

0b. **[packaging track, IN PROGRESS ? 11 PRs merged 2026-06-21/22]** Downloadable batteries-included app ? ``docs/PACKAGING_PLAN.md``. The **LIGHT-tier engine** is bundle-ready: ``pip install`` ? ``indexer --app`` serves the **bundled SPA** single-origin, **torch-free**, with app-home state / ffmpeg resolver / DNS-rebinding guard / single-instance lock / safe DB upgrades (P0a?e, P1a/b, P2-launcher/P2a/P2d/P2e). **Next (fresh focused pushes):** **(1)** P3 **GPU validation + screencast** (L4, ~\$0.50, DELETE-after, \$100 cap ? `[[gcp-vm-always-delete]]``; ? native-WASB 0-rally tuned-config blocker), **(2)** P2b-install (uv/script + truly-local default config), **(3)** P2c first-run wizard (cross-repo khelsutra SPA from ``/api/capabilities``), then P3-cloud (Cloud Run) + P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]`` / `[[next-session-packaging-app]]``.

1. ? **COMPUTE_DECOUPLED_SERVING` M0?M9 LARGELY SHIPPED** (2026-06-21, ADR-014 ratified). M1?M4 (#279/#280/#282/#283), **M5 #286, M6 #287, M8 #288, M9-auth #290** all merged (default-OFF); the **real M7 L4 run** validated the worker (22 rallies, VM deleted, \$0). All \$8 gating Qs locked (D7?D17). **Owner residual** (not blocking Claude's default-OFF code): counsel ToS/DPA (M9-consent live + M11 live), real GCP prices+FX (M8 calibrate), Google OAuth app (M12 real), CF

team-domain/AUD + ingress-lock; real cloud verification authorized within a **\$100/?10k cap** (D17).

2. **[owner, ongoing] Grow the golden corpus** ? multi-court badminton + ?3 matches/sport (TT first). The single highest-leverage move; the quality track *and* full-corpus training feed on it.
3. **[owner + Claude · GPU] Full-corpus training (n?5)** ? a real `weights/<ver>/` (the `0.1.0-l4` bundle is n=2 plumbing-only).

P1 ? near-term, mostly Claude-doable (default-OFF, green-gated)

4. **[Claude] `COMPUTE\_DECOUPLED\_SERVING` batch ? REMAINING: M11 + M12** (M5/M6/M8/M9-auth ? merged this session). **M11** = `backend/flywheel.py` capture?store?shadow-eval?goldenize plumbing reusing `workspace.py::for\_video` + `backend/eval/experiment.compare` + `backend/eval/promotion.py::promote\_if\_served\_safe`, behind `flywheel.\*`=OFF + a faked `consent\_attested` (? privacy-sensitive ? give it fresh focus). **M12** = `backend/storage/gdrive\_api.py` `GDriveApiStorage(StorageBackend)` + a `gdrive-api://` scheme (regex already allows the hyphen; add to `\_KNOWN` + `\_backend\_class` + the StorageConfig Literal) against an INJECTED fake Drive (mirror `gcs.py` DI); keep the mount-only `gdrive://` untouched. ? **M9-consent cols = a new v7 migration** (M8 took v6). M9-consent ToS/DPA text = owner/counsel.
5. **[Claude] Per-video workspace P1?P6** (P1 v6 DB migration ? P2 worker ? P3 per-video routing of `detector/`+`tmp/` *[the `output/chunks\_` de-hardcode base = ? done via #282]* ? P4 ? P5 Launch-Report flip ? P6 GPU parity). ? P1's v6 migration coordinates with the M8 v6 cost-ledger migration ? author the version bump once.
6. ? **Ops-Agent codified (#291)** ? `deploy/gcp/create\_gpu\_vm.sh` (the one-command VM helper) always bakes the Ops-Agent startup-script; provision.sh grants the APIs+SA roles. PROVEN active on a fresh VM 2026-06-21. ([[gcp-ops-agent-enable]])
7. **[owner] Set the ship-gate target (#172)** ? needs corpus growth.
8. **[owner-side GPU] R-series: promote-or-reject `R2` toLF1 as the gate** (the ablation) ? needs GPU-extracted golden features.

P2 ? later / gated / deferred

9. Rally-quality **P3/P4** (person-fusion LOVO, audio cue) ? needs GPU golden features.
10. **M0.4 scorer swap** ? TemporalMaxer (per the 2026-06-17 external review).
11. **Multisport** (table-tennis first, then pickleball).
12. Serving **M8/M9** ? per-video cost ledger, edge auth, DPA/consent (within `COMPUTE\_DECOUPLED\_SERVING`; owner/legal-gated).
13. **WASB-C3** weight provenance (gates *sharing* a trained model) ? legal.
14. **Rename repo off "badminton"*** (owner, BACKLOG).
15. Owner housekeeping: retire the shared **DO droplet** + **DO token**; collector **`md5`** export (collector repo).
16. **PMF field validation** (owner-side, business track).

Doc-hygiene (tracked in `docs/DOC\_STATUS.md` §4): ? #1 per-video reconcile + ? #2 this refresh; **pending:** field/PMF/Roora ? archives + a data-layer `docs/` dir (owner boundary confirm); confirm the COMPLETED top-level stubs; verify `HOSTING\_PLAN`/`UI\_PROPOSAL` currency.

Owned-model & rally-quality track ? detail (2026-06-13 plan; still valid)

Updated: 2026-06-13 (**rally-detection quality track**: fusion + experiment-harness design docs merged, owner-gated ? see the dedicated section below, the desktop-agent pick-up). Prior: 2026-06-10 (audit?M0?Gemini-battery session; later same day the **UI-alpha track opened**:
ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private `khelsutra` repo, local-first delivery ? see
`docs/UI\_ALPHA\_EXECUTION\_PLAN.md`; **PR-1 async jobs (#16) MERGED 2026-06-11, PR #87**; **PR-2 upload/download MERGED 2026-06-11, PR #99** ? live-E2E-verified (3-part chunked upload MD5 byte-identity + compile?browser-download); next: PR-3 identity. Same day: the **8?9 quality sweep #90?#98 landed** ? ruff/mypy/
coverage-floor CI lanes, SQLite conn-close fix, config unknown-key warning, hybrid-twin collapse into
`trajectory\_hybrid.py`, WSL tilde+abspath live-path fixes; suite 388 green; LOVO 0.626 re-

verified exact).

****Authoritative roadmap:****

`docs/OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md` (+ ADR-008 for topology). ****Latest blow-by-blow:****

memory

`current-state.md` / `session-handoff.md`. This file = the prioritized "what to do next,"

refreshed each session.

? Next-session / GPU-box pickup (2026-06-15)

****M0 in-container quick wins are COMPLETE:**** ship-gate provisional blocker (****#170****), train/eval split guard (****#171****), the ****doc-consistency sweep**** (****#174**** ? corrected 16 drifted docs vs the code), and the ****P3 person-feature fusion litmus machinery** is already present + suite-green (``backend/eval/fusion_compare.py`` + ``ablation.py``: person-driven ?F1 / bootstrap CI / paired sign test; the real-golden ``skipif`` litmus waits only on GPU-extracted data). ? CI is red repo-wide due to a ****GitHub Actions minutes/billing outage**** (account-level, not code) ? work is local-gated + ``--admin``-merged until Actions is restored (Settings ? Billing ? Actions minutes / spending limit).

****Immediate next step ? the real P3 LOVO measurement**** (this box has a GPU + ``torch+cu124`` + ``cv2``; videos and trajectory CSVs are present, but the GPU-extracted golden features + ``rallies.csv`` labels are not yet here):

1. ****[owner / GPU box]**** ``python -m backend.eval.fusion_golden --manifest output/human_lovo_manifest.json --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-data\features\2026-06-15-n6"`` ? produces ``*_golden_features.json`` for the 6 golden videos. (GPU + WSL + videos + trajectory CSVs + ``rallies.csv`` labels are owner-side only.)
2. Artifacts land in the ****shared cloud folder**** ``C:\Users\avidu\OneDrive\rally-annotator\golden-data\`` (OneDrive auto-syncs both ways). Convention + workflow: ****[`GOLDEN\_DATA\_SHARING.md`](data_pipeline/GOLDEN_DATA_SHARING.md)****; tracked in ****#175****.
3. ****[CPU dev/agent session]**** ``python -m backend.eval.fusion_compare --features-dir <shared>\features\2026-06-15-n6 --record`` + ``python -m backend.eval.ablation --features-dir <shared>\... --granularity group`` ? real person-feature ?F1; copy the JSONs into ``output/`` to also pass the ``skipif`` litmus tests (``tests/test_ablation.py::test_litmus_reproduces_gate_a``).

****Also queued (owner-gated / pending input):****

- Ship-gate target calibration ? ****#172**** (owner decision; needs corpus growth).
- Tests reorganization ? blocked on the owner's grouping metric (provide it ? an agent will do it).
- Restore CI ? the Actions outage is account-level (billing/minutes), not a repo/config bug.

****Leaving (bulky / M2 ? no start without owner go):**** ActionFormer (M0.4), event-index DB migration (M0.1), Gemini-silver purge (M0.3), cost-to-Gen-2 benchmark + ByteTrack deprecation, multisport, repo rename, PMF execution.

Where we are (one paragraph)

ADR-008 ratified ("repo split driven by productionization, not isolation"): training lives in-repo (``training/``) behind a CI-enforced import wall; the featurizer is single-sourced (``backend/features/rally_seq.py``, `train==serve`); the ****Gen-0 harness ships**** (``python -m training.gen0.harness --manifest output/human_lovo_manifest.json --floor eval_baselines/heuristic_lovo_n6.json --version <semver>``) and produced artifact v0.1.0 (LOVO ****0.626****, floor passed, sign test 5/6 wins `p=0.109` ? honestly not significant, `ship=False`). The ****n=6 owned corpus** is in the collector (262 rallies, Proprietary, commercial export = owned data only after the ShuttleSet reclassification). The ****Gemini battery is done**** (see table) ? the moat is quantified.

The quality table that now drives strategy (IoU 0.5 vs human golden labels)

Path	Mahadevpura (clean-ish)	Kushagra (multi-court)
--- --- ---		
Heuristic (velocity windowing)	0.657	0.157
Gen-0 owned head (LOVO, n=6)	0.744	0.561
Two-stage WASB+Gemini refine	0.83	?
Gemini wrapper (<code>`gemini-flash-latest`</code>)	**0.93**	**0.66**

****Reading:**** clean footage is commoditized (serve it with Gemini for cents). ****Multi-court drops the commodity wrapper to 0.66 ? that is the owned model's ship target**** (its FPs are background-game pickups + dead-time merges, the exact contrast-metric confound). Gen-0 is 0.10 behind with only n=6 ? a corpus-growth-sized gap, not an architecture-sized one. Two-stage refine = the cost-efficient serving path for LONG tapes (uploads candidate clips only); wrapper 503-flakiness (observed all session) makes the provider-fallback registry load-bearing. Baselines tracked in `eval\_baselines/` (heuristic_lovo_n6.json, gemini_wrapper_2026-06-10.json).

Rally-detection quality track (design merged 2026-06-13, owner-gated; implementation in progress)

****This is the track a desktop/local agent is slated to pick up.**** Design docs landed (PR #112 + session). All implementation is ****owner-gated**** (per explicit in the plan); each step = ONE PR off `master`.

****Hardening loop (T1-T5 positive tests + audit + finale) COMPLETE (2026-06-14, PRs #156/#157/#160/#161/#163 + #165 + final records).**** See archived `docs/archives/CODE\_AUDIT\_AND\_TEST\_HARDENING-COMPLETED-2026-06-14.md` (Living Log with every pre/post/review) and `docs/archives/HARDENING\_LOOP\_HANDOFF-COMPLETED-2026-06-14.md` (full rules + STATE). Top-level now pointer stubs per #165 review. All 5 items + verification + archive done; clean slate.

- ****Docs:**** `docs/MULTI\_SIGNAL\_FUSION\_PLAN.md` (shuttle + person + audio, one fusion layer) .
`docs/EXPERIMENT\_HARNESS.md` (A/B + shadow + promotion gate, zero-regression) .
`docs/ROORA\_LABELING\_GUIDELINES.md` (v8) . `docs/HOW\_RALLY\_DETECTION\_WORKS.md` (2-layer mental model).

- ****Key finding (no re-investigation needed):**** the owner's "held-shuttle counted as a rally" bug is ****NOT** a missing capability ? `backend/pipeline/detectors/rally\_gate.py` already implements the net-crossing ****"Gate A" (apply_gate(..., min_crossings=2))** that kills service-feed/held-shuttle windows, but it is ****only called from eval drivers**** (and there was no `min\_crossings` knob in `IndexingConfig`). ****Wiring Gate A into the live segmenter (now via fusion_hybrid) was the single cheapest, highest-confidence quality win.**** (P2 FusionSegmenter specifics: registry `fusion\_hybrid`, ****default-OFF****, seeds from substrate, persists `net\_crossings` + optional person features, optional served Gate A/B via config knobs; adversarially reviewed; suite green. See `tests/test\_served\_gate\_a\_regression.py` for the honest finding that on production windowing Gate A is neutral/negative in some cases, hence kept opt-in with knob for safety, and the leaderboard revert note.)

- ****Current status (2026-06-14):**** P1 (person cue extraction to reusable `PersonDetector`) + P2 (FusionSegmenter + Gate A signal + served handoff gating) + experiment harness P1 DONE (adversarial review + NO-REGRESSION; suite green). ****P3 next (this PR + follow-ups):**** learned fusion (person features ? harness `compare` LOVO lift on the 6 golden videos; eager-person-compute optimisation); then P4 audio via hit_detector.

- ****Pick-up order (each = ONE PR off `master`, owner approval first, flag-gated default-OFF, promote only on measured LOVO):****

1. (P2 first-step, largely done via fusion_hybrid) Wire Gate A into production served path + config knob (min_crossings); apply in hybrids at runtime. (Now available opt-in; served neutral on production windowing per litmus ? kept opt-in with knob for safety.)

2. P3 first step (this work item): harness `compare` LOVO litmus for person-on variant (players_in_court + two_sided + colocation etc. from `fusion\_features`) on the 6 golden; add the eager compute optimisation note/landing. Pure + harness (testable here).

3. P3/P4 continuation: audio cue integration + full learned promotion if lift proven.

- ****Discipline (from EXPERIMENT_HARNESS.md):**** every new cue ****flag-gated, default OFF****; promote to default ****only on measured LOVO lift**** through `run\_regression.py` (exit 0/1/3); pinned `CONTROL` ? rollback = config flip. Harness re-uses ~80% existing code.

- ? Gate A, person cue logic, harness, and pure fusion paths are testable here; the heavier real-video / golden LOVO / ablation confirmation waits on the GPU-extracted golden features + `rallies.csv` labels, which are not yet present in this checkout.

- ****External due-diligence review inputs (2026-06-17) ? PICK UP AFTER current ideas are exhausted; full details + citations in [RALLY_QUALITY_RESEARCH.md] (RALLY_QUALITY_RESEARCH.md) §5.6.**** An owner-commissioned 31-reference review of the quality report ****validated**** the methodology and surfaced these ****future**** levers (research inputs, not active work, not new claims; verify citations before any submission):

1. **"First-mile" paper framing (highest-value):** stroke-classifiers (BST/DSTA) + the big datasets (ShuttleSet ~36k strokes, Fine-Badminton ~27k actions) all **assume a pre-trimmed rally clip already exists** ? we solve the ignored prerequisite. Pitch the paper as **"the missing untrimmed-video preprocessing pipeline that lets BST/MonoTrack run on raw footage."**
2. **Ground R2 in Action-Spotting / Precise-Event-Spotting** (SoccerNet/TennisSet/T-DEED) ? turns R2 into a defensible publishable contribution.
3. **Add domain baselines for any submission:** MonoTrack + BST on ShuttleSet / Fine-Badminton (we have only heuristic + Gemini today).
4. **Prefer TemporalMaxer over ActionFormer** for the M0.4 scorer swap (parameter-free max-pool, ~2.8x fewer GMACs, small-N robust) ? see "M0.4 next increment" below.
5. **Court polygons ? pose ? the rally-END lever:** masking on-court players can resolve **"occluded end-of-rally states"** (the #1 remaining gap vs Gemini), not just spectator noise.
6. **Name the eval/serve lesson "pipeline distribution skew"** ? a citable paper "lesson."

Discipline reminder: owner approval before starting any owner-gated item; one PR per step; babysit the PR (poll/fix/reply) until merge observed, **then** advance.

Prioritized next steps

1. **[owner, in progress] GROW THE GOLDEN CORPUS** ? the single highest-leverage move; every lever below feeds on it.
 Priority order for new videos: **(a) multi-court badminton** (the target distribution AND where the ship target lives), **(b) ?3 matches per new sport ? table tennis first** (WASB transfers at F1 0.909; pickleball labels are corpus-gold but not trainable until a clean MIT/Apache ball detector exists), (c) capture variety per ``VIDEO_RECORDING_GUIDELINES.md``; **adult sessions only for the training pool** (consent guardrail). Per video:
 label ? ``curate import-local --normalize --license Proprietary --fps <true fps>`` ? re-run the Gen-0 harness.
 The paired sign test needs n: at n=10 with 9 wins, $p > 0.011$? significance is reachable this month.
2. **[owner decision] Set the ship-gate target** ? now informed: floor = heuristic 0.480 (necessary), **multi-court**
 reference = wrapper 0.66 (the number that makes the owned model worth serving), clean-footage ceiling = 0.93
 (not the target ? Gemini serves that tier). Suggested shape: "LOVO F1@tIoU0.5 ? 0.70 on multi-court folds with paired-sign $p < 0.05$ vs heuristic" ? owner to ratify/adjust.
3. **M0.4 next increment (\$0/local):** swap the Gen-0 LogReg for a lightweight temporal head (MIT, minutes on the 2070S ? NOT the heavyweight Gen-2 stack) inside the existing harness; per-video threshold calibration (the LOVO-vs-oracle gap is visible per fold in the harness output). **Prefer TemporalMaxer** (parameter-free max-pool, ~2.8x fewer GMACs, small-N robust) over ActionFormer/ASFormer per the 2026-06-17 external review
 (``RALLY_QUALITY_RESEARCH.md`` §5.6) ? better fit for the cheap/local/fast mandate.
4. **Multisport thread:** ingest Roora pickleball/TT CSVs (``import-local --normalize``); merged-prototype LOVO across badminton+TT once TT has ?3 matches; identify a clean pickleball ball detector. ? TT-on-WASB serving stays gated by C3 (provenance multiplies per sport).
5. **PMF validation (cheaper than any engineering month):** C13 academy interviews (+ the two added questions:
 "multi-court footage nobody watches?" / "pay per-match to index it?"); camera pilot at the warmest 2-3 academies
 (consent at capture; demo served via the Gemini path or human-polish ? the reel pattern exists:
``output/MahadevpuraSingles_highlights.mp4``). **Field kit COMPLETE** (2026-06-12, owner ratified the physical-visit approach; reframed product-feedback-first per owner spec):
``docs/PRODUCT_FIELD_ASSOCIATE_FLYER.md`` (the recruiting flyer ? owner fills ?/contact placeholders and recruits) +


```
`docs/FIELD_VISIT_PLAYBOOK.md` (the hired associate's visit playbook: coach + student script,
demo-after-Q6
    rule, ground rules, Phase-2 record-and-show-back stretch goal) +
`docs/FIELD_VISIT_ANSWER_SHEET.md` (per-visit
    capture form incl. demo-reaction + student blocks) + `docs/PMF_FIELD_SIGNALS.md` (G/A/R
anchors + pre-registered
    decision rules PROCEED / PIVOT A-C / FALSIFIED thresholds fixed before the first visit).
6. Housekeeping: rotate the Gemini API key DONE 2026-06-10 (owner-verified working;
confirm old-key
    deletion in AI Studio); rename repo off "badminton" (owner action,
    in BACKLOG); collector=SoR confirmation; off-peak niceties: none pending ? the battery
completed.
```

Pointers / artifacts

```
- Roadmap: `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` · Topology: ADR-008
(`docs/archives/decisions/DECISIONS.md`) ·
    Study: `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md` · Quality history:
    `docs/archives/quality-iterations/2026-06-multivideo-lovo/`.
- Harness: `training/gen0/harness.py` (one command ? train ? LOVO ? gate ? artifact bundle;
weights gitignored,
    manifest reviewable). Featurizer: `backend/features/rally_seq.py` (FEATURE_SCHEMA v1).
- Corpus: collector DB (system-of-record, 6 matches/262 rallies) · eval manifest
`output/human_lovo_manifest.json`
    (gitignored) · durable trajectories + Roora files: `C:\Users\avidu\Projects\Annotation
Setup\Collect\Trajectories\` ·
    labels: `...\Collect\Golden Labelled\`.
- Gemini ops learnings: serial uploads (`indexing.ai_max_workers=1`) + chunk re-encode
    (`indexing.ai_chunk_scale_width=1280`) + 6-attempt exponential generate retry ? a cold/prepaid
project burst-throttles
    with a MISLEADING "credits depleted" 429; ramp gradually.
- ? No owned-model implementation beyond greenlit M0.x until owner greenlights the next
milestone. Committed next
    PLATFORM slice = async job model (single-tenant).
```

8. assistant (2026-06-22T01:56:14.528Z)

Now let me read DOC_STATUS.md and BACKLOG.md, plus check the Finding B specifics ? how the gemini segmenter and job status handle the missing-key/0-rally case.

9. user (2026-06-22T01:56:15.574Z)

Documentation Status & Archival Register

```
> Status: ? LIVING ? the single board for doc lifecycle + archival due-diligence.
Last full review: 2026-06-21.
> Update cadence: whenever a doc changes lifecycle state (created / shipped a milestone /
delivered / archived), and re-review at each clean-slate.
> Why this exists: so we can (a) review completeness at a glance, (b) update it as
work finishes to prove we're done, and (c) never archive a foundational doc without honestly
updating it first (owner directive, 2026-06-21).
```

0. How to use this

```
- Reviewing now? Read §4 (the open findings/actions) ? that's what needs attention. §3 is
the full register.
- Marking something complete? Update the doc's own §7 progress tracker (the source of
truth for its completeness), then flip its row here, then ? if terminal ? run the §2 archival
checklist.
- This register does NOT duplicate completeness detail ? it indexes lifecycle + archive-
readiness and points to each tracked doc's §7.
```

1. Lifecycle legend

`DRAFT ? IN PROGRESS ? DONE/DELIVERED ? ARCHIVED`. Plus: ****LIVING**** (permanent reference, never "done"), ****PROPOSAL/DESIGN**** (owner-gated, not started), ****REPORT**** (a terminal record of a run/launch), ****CONVENTION**** (a standing rule/guide).

2. Archival due-diligence checklist ? *the repeatable process (do NOT just `git mv`)*

Before a foundational/tracked doc moves to `docs/archives/`:

1. ****VERIFY** claims against the code ? no stale assertions; tag load-bearing facts `[verified]`. (A doc that says "nothing built" when code shipped is **not** archive-ready ? fix it first.)
2. ****Flip Status** ? DELIVERED / DONE / REJECTED + a completion note ? what shipped, where (PRs, paths, artifacts), and what was ****deferred** / carried forward
3. ****Update ALL inbound references**** ? `docs/README.md` index, `CLAUDE.md`, `docs/NEXT\_STEPS.md`, and any cross-doc links ? the new path + status.
4. ****Keep the lineage**** ? ADR forward/back-links + "supersedes / superseded-by" so the evolution stays traceable.
5. ****`git mv` the whole dir/file** ? `docs/archives/past\_projects/` (terminal-state ONLY; live docs stay at `docs/` top level). Relative links inside become historical (note this in the doc's header).
6. ****Record the move**** ? flip the row here to §3e + a line in memory `current-state`.

> ****Worked example (the template):**** ****DECOUPLED_COMPUTE ? #270**** ? verified M0?M2+M3.5 delivered on GCP, flipped ??? DELIVERED with a completion note (incl. carried-forward WASB-C3/DPA/md5), reindexed `docs/README` + `past\_projects/README`, kept the ADR-010?014 lineage, then `git mv` to `docs/archives/past\_projects/DECOUPLED\_COMPUTE/`.

3. The register

3a. Tracked projects ? lifecycle docs (archive when DONE; each doc's §7 is the completeness source-of-truth)

Doc	Status	Archive-ready?	Next action
--- --- --- ---			
`COMPUTE_DECOUPLED_SERVING/` **IN PROGRESS** ? M0 approved + **M1?M4 SHIPPED** (#279/#280/#282/#283); M5+ default-OFF code batch authorized (2026-06-21) No M5 truly-local ? M6 Cloud Run ? M8 ledger ? M9-auth ? M11 flywheel ? M12 gdrive-api (owner residual: counsel/prices/OAuth/real-run within the D17 cap)			
`RALLY_QUALITY_RESEARCH.md` Foundation, **tracked & ACTIVE** (§7) No Drives the R-series; complete when §7.1 single quality number is attributed			
`R2_EVAL_METRIC_DESIGN.md` IMPLEMENTED (augmenting) No (gates R4 + dual-eval; ablation pending) Promote-or-reject to f1 as the gate (ablation) ? then archive with the R-cluster			
`R1_MILESTONE_LAUNCH_REPORT.md` LAUNCHED (#204) ? terminal record Soon (with the R-cluster) Keep with R-series for narrative; archive when the track wraps			
`RALLY_DETECTION_GAP_CLOSURE.md` DRAFT (owner-gated) No G1?G3 levers; part of the R/quality cluster			
`PER_VIDEO_WORKSPACE.md` DRAFT, IN PROGRESS (P0 = #264) No ? supersede-reconcile (finding #1) DONE; P3 de-hardcode base ? via #282; remaining P1?P6 (P1 v6 migration coordinates with M8's v6 cost-ledger ? version bump authored once)			
`MULTI_SIGNAL_FUSION_PLAN.md` P1 DONE; P2+ owner-gated No P3/P4 (learned fusion, audio) ? needs GPU golden features			
`EXPERIMENT_HARNESS.md` P1 IMPLEMENTED No A/B + promotion gate; supports the quality track			
`PROMINENT_COURT_DETECTION.md` DRAFT (owner-gated) No C-series; multi-track selection design			
`EXECUTION_POLICY.md` P1 SHIPPED; P3b next No Async job policy; reused by Cloud Run dispatch			
`GOLDEN_REGRESSION_FIXTURES.md` IN PROGRESS (owner-gated) No Golden eval fixtures			
Data layer ? `docs/data_pipeline/` (`GOLDEN_SET_IMPLEMENTATION` . `GOLDEN_SET_PHASE1_VIDEOS` . `GOLDEN_VIDEOS` . `VIDEO_RECORDING_GUIDELINES` . `GOLDEN_DATA_SHARING`) LIVING (data, **no-ML**) n/a Relocated 2026-06-21 to the isolated data pillar; tracked via `data_pipeline/README.md`. Roora <i>*vendor*</i> docs archived ? `archives/roora-vendor/`.			
`DESKTOP_APP_PLAN.md` DRAFT (owner-gated, nothing built) No P0 compute-feasibility spike; relates to truly-local profile			
`OWNED_MODEL_IMPLEMENTATION_PLAN.md` PLAN/roadmap (owner-gated) No (authoritative roadmap) Gen-0?Gen-2; the owned-model track			
`PERSONALIZATION_PLAN.md` DESIGN ? owner-adopted; Phase-0 No Per-user models (DB v5			

```
store landed) |
| `UI_ALPHA_EXECUTION_PLAN.md` | ACTIVE | No | khelsutra UI alpha track |
| `I18N_PLAN.md` | Design (P0); **in implementation** (adapted) | No | Executed in
`khelsutra/docs/FRIENDLY_AND_MULTILINGUAL.md` (P1?site/, P3?web/); P2 backend stays here, owner-
gated |
| `POST_166...RISK_ASSESSMENT.md` | Plan for owner review | No | Milestones/risk doc |
```

3b. Foundational / strategy references (LIVING ? archive only if ****superseded****, not when "done")

```
| Doc | Status | Note |
|---|---|---|
| `PLATFORM_ARCHITECTURE.md` | Strategy/research | The `ComputeBackend`/`StorageBackend`
registries ? realized by COMPUTE_DECOUPLED_SERVING |
| `COMMERCIALIZATION.md` | ? LIVING | Strategy; carries WASB-C3 |
| `VIDEO_LOCALITY_MODEL.md` | THINKING DOC | L0 fully-local = the truly-local profile |
| `HOSTING_PLAN.md` | PROPOSAL | **? partly overtaken by ADR-013 + the shipped site ? verify
(finding #5)** |
| `UI_PROPOSAL.md` | PROPOSAL (partly shipped) | UX half **in implementation** ?
`khelsutra/docs/FRIENDLY_AND_MULTILINGUAL.md`; currency banner added (finding #5 ? for this doc)
|
| `STORAGE_SHARING_MODEL.md` | DRAFT (co-design pending) | ? |
| `ANALYZERS_AND_RECONCILERS.md` | DESIGN (owner-gated) | ? |
| `ROADMAP.md` / `CODE_MAP.md` / `HOW_RALLY_DETECTION_WORKS.md` /
`KHELSUTRA_PRODUCT_OVERVIEW.md` | LIVING references | Keep current; not archive targets |
```

3c. Reports & terminal records (archive candidates once their track wraps)

```
| Doc | Status | Note |
|---|---|---|
| `REAL_FOOTAGE_VALIDATION_REPORT.md` | validation complete | Motivates R2; archive with the
R-cluster |
| `RALLY_DETECTION_QUALITY_REPORT.md` | report | Quality snapshot; part of R-cluster |
| `QUALITY_ITERATIONS.md` | LIVING empirical log | **Keep** ? the paper-backbone ablation log
([[paper-coauthor-aim]]) |
| `CODE_AUDIT_AND_TEST_HARDENING.md` . `HARDENING_LOOP_HANDOFF.md` .
`DEFERRED_HARDENING_PLAN.md` | COMPLETED ? **top-level pointer stubs** (archived copies exist) |
**? confirm intentional stubs (per #165) vs finish-archiving (finding #4)** |
```

3d. Conventions / guidelines / templates / runbooks (permanent ? do not archive)

```
`PROJECT_DOC_TEMPLATE.md` . `NEXT_STEPS.md` (? refreshed 2026-06-21) . `BACKLOG.md` .
`EVALUATION.md` . `ABLATION_LAYER.md` . `SETUP_NEW_MACHINE.md` . `TRACKNET_WSL_SETUP.md` .
`README.md`. *(2026-06-21 relocation: the data-layer guides moved ? `data_pipeline/`; the field-
ops + Roora docs ? `archives/{field-pmf,roora-vendor}/`.)*
```

3e. Already archived (`docs/archives/`)

```
`past_projects/DECOUPLED_COMPUTE/` (? 2026-06-21, #270) . `past_projects/low-regret-moves/` .
**field-pmf/`** (field-ops + customer-feedback, 2026-06-21) . **`roora-vendor/`** (annotation-
vendor engagement, 2026-06-21) . `PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md` .
`CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md` . `DEFERRED_HARDENING_PLAN-
COMPLETED-2026-06-14.md` . `HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md` .
`COMMERCIAL_READINESS_REVIEW.md` . `MONETIZATION_AUDIT.md` . `TEST_RUN_SUMMARY.md` . `research/`
. `checkpoints/` . `quality-iterations/`.
```

3f. ADR log ? `docs/archives/decisions/DECISIONS.md`

```
**ADR-001 ? ADR-014** (14, all ratified). The compute/serving lineage: **ADR-005 ? 008 ? 009 ?
010 ? 011 ? 012 ? 013 ? 014**. ADR-014 was **RATIFIED 2026-06-21** (owner M0 sign-off for
COMPUTE_DECOUPLED_SERVING).
```

4. Open due-diligence findings / actions (2026-06-21 review)

```
> **? Doc-restructuring done (2026-06-21, owner-directed):** field-ops/customer-feedback ?
`archives/field-pmf/`, Roora annotation-vendor ? `archives/roora-vendor/`, and a no-ML **data
layer** carved into `docs/data_pipeline/`. Inbound + outbound links rewired; three dir READMEs
added (the `data_pipeline` one sets the "no segmentation/stitching" boundary).
1. **? RESOLVED (2026-06-21) ? `PER_VIDEO_WORKSPACE.md` supersedes
`PER_VIDEO_OUTPUT_LAYOUT.md`.** Folded forward the `video_slug` naming option (workspace $8.5) +
```

the `output/GX010125\_run/` manual-prototype precedent; predecessor archived ?
 `docs/archives/PER\_VIDEO\_OUTPUT\_LAYOUT-SUPERSEDED-2026-06-21.md`; `CODE\_MAP.md` repointed.

2. **RESOLVED (2026-06-21)** ? `NEXT\_STEPS.md` refreshed** with a current 2026-06-21 state + a **cross-track prioritized pending list**; the 2026-06-13 owned-model/quality body is kept (clearly marked) as that track's detail.

3. **R-series / quality cluster is ACTIVE** (`R1`, `R2`, `RALLY\_DETECTION\_GAP\_CLOSURE`, `RALLY\_QUALITY\_RESEARCH`, `QUALITY\_ITERATIONS`, `REAL\_FOOTAGE\_VALIDATION\_REPORT`, `RALLY\_DETECTION\_QUALITY\_REPORT`). **Do NOT archive yet** ? archive as a **cluster** once R2 is promoted-or-rejected as the gate + \$7.1's single number is attributed. (Owner already confirmed "leave as is", 2026-06-21.)

4. **Confirm the COMPLETED top-level stubs** (`CODE\_AUDIT\_AND\_TEST\_HARDENING`, `HARDENING\_LOOP\_HANDOFF`, `DEFERRED\_HARDENING\_PLAN`) are intentional pointer-stubs (per #165) ? or finish-archive them so there's one copy.

5. **HOSTING_PLAN.md** / **UI_PROPOSAL.md** are PROPOSAL but partly overtaken** by ADR-013 (DO dropped) + the shipped khelsutra site. **UI_PROPOSAL.md** banner added (2026-06-21)** ? its UX half is now tracked in `khelsutra/docs/FRIENDLY\_AND\_MULTILINGUAL.md`; the **I18N_PLAN.md** localization track is likewise in implementation there. **HOSTING_PLAN.md** banner still pending** (ADR-013 already noted in it).

6. **RESOLVED (2026-06-21)** ? **COMPUTE_DECOUPLED_SERVING** M0 approved + M1?M4 SHIPPED** (#279/#280/#282/#283); all \$8 gating Qs locked (D7?D17); M5+ default-OFF code batch authorized (owner residual: counsel ToS/DPA, real GCP prices, OAuth app, real cloud run within the D17 \$100/?10k cap).

5. Maintenance

Update a row when its doc changes lifecycle; run \$2 before any archive; re-review \$4 at each clean-slate. This register + the \$2 checklist are the operational form of the `CLAUDE.md` archive policy and [[working-agreement-tracked-project-docs]].

10. assistant (2026-06-22T01:56:18.998Z)

Now let me look at Finding B specifically ? the gemini segmenter and job-status handling for the 0-rally / missing-key case. Let me examine the relevant backend code on origin/master.

11. user (2026-06-22T01:56:22.014Z)

```
=== gemini segmenter files ===
backend/eval/gemini_refine.py
backend/pipeline/segmenters/__init__.py
backend/pipeline/segmenters/_keyhint.py
backend/pipeline/segmenters/base.py
backend/pipeline/segmenters/fusion_hybrid.py
backend/pipeline/segmenters/gemini.py
backend/pipeline/segmenters/motion.py
backend/pipeline/segmenters/tracknet_hybrid.py
backend/pipeline/segmenters/trajectory_hybrid.py
backend/pipeline/segmenters/wasb_hybrid.py
backend/pipeline/segmenters/yolo_hybrid.py
backend/providers/gemini.py
eval_baselines/gemini_wrapper_2026-06-10.json
scratch/gemini_video_debugger.py
scratch/test_ai_segmenter.py
tests/test_gemini.py
tests/test_gemini_refine.py
tests/test_motion_segmenter.py
tests/test_provider_gemini.py
=== job/jobs API ===
backend/api/job_worker.py
tests/test_async_jobs.py
tests/test_job_self_scheduling.py
tests/test_job_worker_loop.py
tests/test_job_worker_module_seam.py
```

12. user (2026-06-22T01:56:26.386Z)

```
=== job_worker.py: how it sets job status on success/failure ===
```

```

5:that raises `JobWaiting` is parked (status 'waiting' + `next_poll_at` + progress) and
28:# parallel uploads (2026-06-10 ops learning: serial uploads + backoff). The segmenters
46: is the segmenter's `_execute_processing`, run UNDER the destroy-after-confirm contract.
48: The wiring is additive: the production segmenter path never raises this today (zero
50: (resumable AI handoff) raises once a long op exhausts its in-process wait budget.
54:     super().__init__(f"job parked for {delay_sec:.0f}s")
55:     self.delay_sec = max(0.0, float(delay_sec))
68:     segmenter=job.get("segmenter"),
73:     """Run ONE job already claimed (`status='running'`) by `claim_due_job`, recording its
76: Job `status` is the EXECUTION state (queued|waiting|running|done|failed): 'done' means
78: result["status"]; 'failed' means the worker itself crashed/raised; 'waiting' means the
90:     db_instance.mark_job_done(job_id, result)
92: except _main.JobWaiting as w:
94:     logger.info(f"Job {job_id} parked for {w.delay_sec:.0f}s (resume when due).")
96: except Exception as e:
97:     logger.error(f"Job {job_id} crashed: {e}\n{traceback.format_exc()}")
98:     db_instance.mark_job_failed(job_id, f"{type(e).__name__}: {e}")
108: ``billing.enabled=True`` AND a real calibrated pricebook, every recorded cost is **0**
until a
111: With the placeholder pricebook the cost is 0 regardless.
128: except Exception as e: # never wedge a completed job on bookkeeping
129:     logger.warning("Cost recording for job %s failed (non-fatal): %s", job.get("id"), e)
136: Each parked job (`JobWaiting` ? status 'waiting') drops out of this drain and is picked
142: ran = 0
151: # on the next submit/drain. Best-effort ? never raises into the worker.
163: except Exception as e: # pragma: no cover - defensive; never wedge the worker
164:     logger.error(f"Could not compute next-due delay: {e}")
169: _main._arm_wake_timer(max(0.0, min(delay, _main._MAX_DRAIN_WAKE_SEC)))
184:_MAX_DRAIN_WAKE_SEC = 60.0

```

13. assistant (2026-06-22T01:56:30.296Z)

Key detail confirmed: at line 76-78, ``done`` means the segmenter ran to completion (result["status"]), and ``failed`` only means the worker itself crashed/raised. So a Gemini segmenter that swallows the missing-key error and returns 0 rallies would record ``status='done'``. Let me check the gemini segmenter and provider to confirm how the missing key is handled.

14. user (2026-06-22T01:56:32.065Z)

```

=== providers/gemini.py: key handling ===
13:logger = logging.getLogger(__name__)
20: def __init__(self, api_key: str, model_name: str):
21:     super().__init__(api_key, model_name)
23:     self._client = None
27:     if self._client is None:
28:         self._client = genai.Client(api_key=self.api_key)
31: def _safe_delete(self, video_file) -> None:
32:     """Best-effort cleanup of an uploaded File (never raises)."""
35:     except Exception:
52:     except OSError:
55:         logger.error(f"[{log_prefix}] {video_path} is {size_mb:.0f}MB > {MAX_UPLOAD_MB}MB
cap ? "
59:         return [], metrics
62:         logger.info(f"[{log_prefix}] Uploading {video_path}...")
64:         video_file = None
69:         logger.info(f"[{log_prefix}] Upload complete in
{metrics['upload_time']:.1f}s. URI: {video_file.uri}")
71:         except Exception as e:
72:             logger.warning(f"[{log_prefix}] Upload attempt {attempt + 1}/3 failed: {e}")
75:         if video_file is None:
76:             logger.error(f"[{log_prefix}] Upload failed after 3 attempts.")
77:             return [], metrics
87:             return None if video_file.state.name == "PROCESSING" else video_file
91:                 label=f"[{log_prefix}] processing", logger=logger)

```

```

92:         except PolicyTimeout:
93:             logger.error(f"[{log_prefix}] Processing exceeded
{process_policy.max_wait_sec:.0f}s ? giving up.")
95:             return [], metrics
96:         except Exception as e:
97:             logger.error(f"[{log_prefix}] Status check failed: {e}")
99:             return [], metrics
101:            logger.error(f"[{log_prefix}] Video processing failed:
{video_file.error.message}")
103:            return [], metrics
106:            logger.info(f"[{log_prefix}] Processing finished in
{metrics['process_time']:.1f}s.")
109:            # missing: a generate() that never returns previously blocked forever) + retry the
503/network
111:            logger.info(f"[{log_prefix}] Submitting multimodal request to {self.model_name}...")
114:            raw_text = None
125:            response = run_bounded(_generate, gen_policy, label=f"{log_prefix} generate",
logger=logger)
128:            logger.info(f"[{log_prefix}] Response received in
{metrics['gen_time']:.1f}s:\n{raw_text}")
129:            except Exception as e:                # PolicyTimeout (hang) or the last 503 ?
both end here
130:            logger.error(f"[{log_prefix}] generate gave up after retries: {e}")
138:            return [], metrics
144:            logger.error(f"[{log_prefix}] Response is not a JSON array.")

```

15. assistant (2026-06-22T01:56:37.124Z)

This confirms the pattern: the gemini provider returns `[], metrics` (empty list) on all failure paths and logs an error, but does not raise. Let me check where the API key is validated/loaded and whether a missing key raises or silently produces an empty client. Let me look at how the segmenter/config validates the key, and whether there's a preflight.

16. user (2026-06-22T01:56:40.114Z)

```

=== segmenters/gemini.py ===
12:from backend.providers import provider_registry
15:logger = logging.getLogger(__name__)
27:    return os.path.join(base, "chunks", video_id)
36:        return float(val)
40:            # Gemini returned MM:SS or HH:MM:SS despite being asked for decimal seconds.
48:            except ValueError:
50:                logger.warning(f"Timestamp '{val}' was in colon-separated format (MM:SS/HH:MM:SS)
instead of decimal seconds. Converted to {total:.2f}s.")
51:                return total
53:                return float(val)
54:            except ValueError:
55:                logger.warning(f"Could not parse timestamp string '{val}' as a number. Defaulting
to 0.0.")
56:                return 0.0
57:            logger.warning(f"Unexpected timestamp type {type(val).__name__} for value '{val}'.
Defaulting to 0.0.")
58:            return 0.0
64:            return "gemini"
68:            return "High-precision AI-powered semantic play segment detection using Google Gemini
Multimodal API."
70:    def _resolve_provider(self, model_name: str):
71:        """Resolve the sport profile, provider, and API key (the process_video preamble).
73:        Returns (sport_profile, provider) on success, or None to signal any of the
74:        early-return cases (unknown sport, missing API key, unknown provider) ? the
75:        caller returns [] for None.
81:        except KeyError as e:
82:            logger.error(str(e))
83:            return None
85:        # Get AI provider and model config

```

```

87:         provider_name = self.config.get("ai_provider", idx_cfg.get("ai_provider", "gemini"))
89:         # Get appropriate API key based on the provider
90:         api_key_env_var = f"{provider_name.upper()}_API_KEY"
91:         api_key = os.environ.get(api_key_env_var)
92:         if not api_key:
93:             from backend.pipeline.segmenters._keyhint import api_key_hint
94:             logger.error(
95:                 f"{api_key_env_var} environment variable is not set! "
96:                 f"To run the {provider_name} segmenter, set your API key. "
97:                 + api_key_hint(api_key_env_var)
98:             )
99:             return None
102:         provider = provider_registry.get(provider_name, api_key=api_key,
model_name=model_name)
103:     except KeyError as e:
104:         logger.error(str(e))
105:         return None

```

=== grep GEMINI_API_KEY across origin/master ===

17. assistant (2026-06-22T01:56:44.727Z)

This is the crux of Finding B. At `backend/pipeline/segmenters/gemini.py:92-99`, when `GEMINI\_API\_KEY` is missing, the segmenter logs an error and `\_resolve\_provider` returns `None`, the caller returns `[]` (0 rallies). The job then records `status='done'` with 0 rallies ? the error lives only in the log, never surfacing in the job status. Let me confirm the caller returns `[]` and check whether any tracking doc/issue exists for this. Let me also verify the exposure-safety preflight (#294) doesn't already cover this.

18. user (2026-06-22T01:56:46.368Z)

=== caller of _resolve_provider ===

```

        return sport_profile, provider

def _maybe_trim_video(self, video_id: str, video_path: str) -> Tuple[str, Callable[[],
None]]:
    """Optionally trim the video to indexing.debug_duration seconds (debug feature).

    Returns (video_to_upload, cleanup_temp_files) where cleanup_temp_files removes any
    temp trimmed file created here.
    """
    # A: Check for debug_duration to trim the video first
    debug_duration = self.config.get("indexing", {}).get("debug_duration")
    video_to_upload = video_path
    is_trimmed = False
    temp_trimmed_path = None

    if debug_duration:
        logger.info(f"DEBUG FEATURE: Trimming video to first {debug_duration} seconds for
fast processing...")
        base = output_base(self.config)
        temp_trimmed_path = os.path.join(base, f"temp_trimmed_{video_id}.mp4")
        os.makedirs(base, exist_ok=True)
        if os.path.exists(temp_trimmed_path):
            try:
                os.remove(temp_trimmed_path)
            except Exception:
                pass

    cmd = [
        ffmpeg_bin(), "-y",
        "-ss", "0",
        "-t", str(debug_duration),
        "-i", video_path,
        "-c", "copy",

```

```

        temp_trimmed_path
    ]
    logger.debug(f"Running command: {' '.join(cmd)}")
    try:
        subprocess.run(cmd, check=True, stdout=subprocess.DEVNULL,
stderr=subprocess.DEVNULL)
        logger.info(f"Trimmed video successfully created at: {temp_trimmed_path}")
        video_to_upload = temp_trimmed_path
        is_trimmed = True
    except Exception as e:
        logger.warning(f"Failed to trim video using ffmpeg: {e}. Falling back to full
video.")

    # Helper to clean up local temp files
    def cleanup_temp_files():
        if is_trimmed and temp_trimmed_path and os.path.exists(temp_trimmed_path):
            try:
                os.remove(temp_trimmed_path)
            except Exception:
                pass

    return video_to_upload, cleanup_temp_files

def _process_chunked(
    self,

```

19. user (2026-06-22T01:56:50.561Z)

```

=== process_video entry: how None from _resolve_provider becomes [] ===
70:     def _resolve_provider(self, model_name: str):
200:         scale_width=ai_scale_width if ai_scale_width > 0 else None,
478:         if shot_count is None:
506:     def process_video(self, video_id: str, video_path: str, player_pool: List[str] = None,
max_frames: int = None) -> List[Dict[str, Any]]:
514:         resolved = self._resolve_provider(model_name)
515:         if resolved is None:
516:             return []
554:         return []

```

20. assistant (2026-06-22T01:56:54.280Z)

Confirmed. `process\_video` returns `[]` (line 516) when the key is missing ? a successful "0 rallies" outcome, not a failure. Now let me verify whether the exposure-safety preflight (#294, current working branch) or any existing doc/issue tracks this "0-rally vs failed job" gap. Let me search for tracking of this and check the exposure-safety preflight scope.

21. user (2026-06-22T01:56:58.079Z)

```

=== grep for any doc tracking 0-rally / missing-key job-status surfacing ===
origin/master:docs/CLOUD_GPU_DRYRUN_GUIDE.md
origin/master:docs/CODE_MAP.md
origin/master:docs/COMPUTE_DECOUPLED_SERVING/README.md
origin/master:docs/NEXT_STEPS.md
origin/master:docs/PACKAGING_PLAN.md
origin/master:docs/QUALITY_ITERATIONS.md
origin/master:docs/R2_EVAL_METRIC_DESIGN.md
origin/master:docs/archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md
origin/master:docs/archives/quality-iterations/2026-06-human-gt-tuning/README.md
origin/master:docs/archives/quality-iterations/2026-06-multivideo-lovo/README.md
--- 4957b72 worker 0-segment commit ---
commit 4957b7224ba23c5a39e988b6bc33835b92e4f970
Author: Avi <avi.dullu@gmail.com>
Date: Sun Jun 21 06:51:26 2026 +0530

```


fix(worker): configure logging + explain a 0-segment infer (the cold-start blind spot) (#271)

A cold-start `worker infer` that produced 0 rallies gave NOTHING to analyse ? because the worker CLI never configured logging. The pipeline already logs richly (detector choice, "Running native WASB inference", availability failures via logger.info/error), but at the default root level (WARNING, no handler) every line was silently dropped. backend.main configures logging; the worker didn't.

- `\_setup\_logging()` at the CLI entry (basicConfig INFO, format mirrors backend.main) + a global `-v/--verbose` flag for DEBUG (the native detector command, frame counts). `force=True` so it reliably owns the level even if a library pre-configured logging at import.
- cmd_infer now emits a LOUD, explanatory WARNING on a 0-segment success: video_id + segmenter + detector_impl + the likely causes (native runner unavailable / WASB produced no detections / input resolution ? tuned proxy res) + "re-run with -v". A 0-rally run is no longer unexplained.

22. assistant (2026-06-22T01:57:02.824Z)

Important ? there's a closely related precedent (#271) for the worker CLI: it added a LOUD WARNING on 0-segment success but only in logs, and only for the WASB worker path, not the Gemini path and not the job-status surface. Let me check the CODE_MAP and the NEXT_STEPS mentions of this, plus the exposure-safety preflight to be sure Finding B isn't already covered.

23. user (2026-06-22T01:57:05.810Z)

=== CODE_MAP mention of 0-rally / missing key ===
origin/master:docs/CODE_MAP.md:7:? **2026-06-10 audit-hardening sweep changed several behaviors documented below ? trust git history over stale lines until the next full regen:** `add\_video` is now an **UPsert** (no FK-cascade wipe on re-ingest; segmentation is destroy-AFTER-confirm via `segment\_watermarks`/`prune\_segments`); detector artifacts are keyed by **video_id**, not basename; **`indexing.wasb.\*` is now a real typed config** (was silently dropped); `timeout\_sec` defaults **1800s** (was 0/hang-forever); SQLite opens with **WAL + busy_timeout**;
`match\_intervals` requires real overlap (iou>0); `run\_regression` baseline lives in tracked `eval\_baselines/` and **exits 3 on a missing baseline**;

the API validates `video\_path`/`output\_filename`. New tests: `test\_api\_endpoints`, `test\_compiler`, `test\_wasb\_runner`, `test\_reingest\_safety`, `test\_detector\_keying`, `test\_config\_wasb`, `test\_path\_safety`, `test\_reliability\_hardening` (+ collector `test\_licensing\_gate`, `test\_import\_guard`, `test\_cvat\_robustness`).

origin/master:docs/CODE_MAP.md:179:- `@backend/eval/gemini\_refine.py` ? refine WASB candidates with Gemini over SHORT padded clips (precision lever; eval/tuning, NOT the live pipeline).
`::merge\_overlaps(gap\_tol=0.0)` (reused by distill_local with gap_tol=1.0), `::refine\_window` (? FRESH genai client per worker ? shared throws socket errors), `::full\_video\_rallies`, `::run`/`::main` (`--full-video` skips `--trajectory`). ? requires `GEMINI\_API\_KEY` (raises SystemExit); imports `BASELINE` from calibrate_wasb +
`TUNED`/`write\_segments\_csv`/`seconds\_to\_mms` from export_wasb_segments.

origin/master:docs/CODE_MAP.md:324:| Refine WASB candidates with Gemini (tuning) |
`@backend/eval/gemini\_refine.py::main` (`--full-video` skips `--trajectory`; needs `GEMINI\_API\_KEY`) |

=== exposure-safety preflight scope (working tree HEAD commit 6fdd1d4) ===
commit 6fdd1d4e89df4cbfafd64ac06769ad2639d29c37
Author: Avi Dullu <avi.dullu@gmail.com>
Date: Sun Jun 21 20:38:52 2026 +0530

feat(serving): exposure-safety boot posture ? fail fast on a half-configured exposed engine (default-OFF)

M9 hardening for the alpha-shareable "tunnel-to-a-box" topology. When the owner exposes the engine behind the Cloudflare-Access gate (flips `security.edge\_auth\_enabled` on), the gate is only half the safety story ? a half-configured exposure starts looking healthy then leaks / 500s per request. Verify the rest of the posture at BOOT (server only, via a new `include\_exposure=True` from `\_enforce\_startup\_or\_exit`):

- EXPOSED_NO_VIDEO_ROOT (fatal): `allowed\_video\_root` jail unset ? /api/process could read an arbitrary local file.
- EXPOSED_EDGE_AUTH_INCOMPLETE (fatal): `cf\_team\_domain`/`cf\_access\_aud` missing ? lifts the current first-request 401 to boot.
- EXPOSED_AUTH_EXTRA_MISSING (warn): PyJWT[crypto] not importable ? would 500 per request.

DEFAULT-OFF: `edge\_auth\_enabled=False` (today's default) ? strict no-op; the worker never opts in (it reads --video-uri, not untrusted tenant paths). Byte-identical to today's deployments.

Also aligns `auth.resolve\_tenant` to `.strip()` team/AUD so "configured" means the same at boot and at request time (folded from the adversarial review wf_35b735a9, which confirmed default-OFF byte-identical + no bypass). +15 startup tests + a whitespace-config edge-auth test. Tracker + changelog updated in COMPUTE_DECOUPLED_SERVING/README.md.

24. assistant (2026-06-22T01:57:11.583Z)

Good ? the exposure-safety preflight is purely about the EXPOSED-server boot posture (auth/jail config), not about Gemini runtime key absence or 0-rally job status. So Finding B is genuinely uncovered. The CODE_MAP note at line 179 confirms the eval `gemini\_refine.py` DOES raise SystemExit on missing key ? but the LIVE segmenter path (`segmenters/gemini.py`) does NOT; it returns `[]`. That's the asymmetry.

Now let me verify the alpha-deploy follow-ups: engine #301 retention sweep scheduling, native-speaker review, and monitoring of the live droplet. Let me check khelsutra docs and the broad CF token issue.

25. user (2026-06-22T01:57:14.181Z)

=== engine #301 ? what is it? ===

=== T11 retention #299 ? does it self-schedule or need cron? ===

commit cb7394bafc9ee0d91c7f1d1cc68bb869553f43f3

Author: Avi <avi.dullu@gmail.com>

Date: Sun Jun 21 23:00:08 2026 +0530

feat(serving): T11 ? retention sweep for stale uploads + caches (#299)

Before non-owner (alpha) exposure, uploaded source videos must not linger indefinitely. Extend tools/cleanup_caches.py to sweep the upload landing zone (security.upload_dir, default output/uploads): purge uploaded source videos and the orphaned .partial-<id> temp files aborted uploads leave behind, past a privacy retention window (--uploads-retention-days, default 7d), while reels / DB / CSVs stay durable per the cache-hygiene policy.

- New "uploads" scan location + classify() rules (uploaded source / partial upload; a future per-user subdir is left untouched).
- scan_windows() skips the nested uploads dir so it isn't double-counted as an opaque "other dir" (which silently shielded it from retention).
- plan_deletions() gains an independent uploads_retention_days window (privacy), orthogonal to the cache --older-than guard (disk). Window of 0 = off.
- Loud structured output: a retention-window banner, per-category subtotals, per-item PURGED/FAILED lines on --apply, and a --json plan for a cron/job

wrapper. Still dry-run by default; --apply required to delete.

Window (uploads 7d; reels kept durable) confirmed by owner 2026-06-21.
Unit tests cover the new classify/plan paths + the nested-dir skip.

26. assistant (2026-06-22T01:57:19.119Z)

#301 isn't a commit in engine ? it's a tracking issue/ask. The T11 commit (#299) says the tool is "still dry-run by default; --apply required" and emits "--json plan for a cron/job wrapper" ? i.e. the sweep tool ships but the SCHEDULING (cron/job on the live droplet) is NOT yet wired. The khelsutra commit `bea8a2a` confirms: "T11 retention sweep... tool shipped (engine #299), scheduling tracked (#301)". Let me confirm #301 is a khelsutra issue and find the broad CF token reference.

27. user (2026-06-22T01:57:22.425Z)

```
=== #301 ref in khelsutra ===
docs/LOCAL_APPLIANCE_ARCHITECTURE.md:130:| Retention | sweep **tool** now exists ([engine
#299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299), dry-run-first, uploads
>7d, reels durable) but **not yet auto-scheduled** ([engine
#301](https://github.com/Khelsutra/badminton-highlight-indexer/issues/301)) | run the sweep
`--apply` before/with any non-owner exposure; auto-schedule per
[#301](https://github.com/Khelsutra/badminton-highlight-indexer/issues/301); keep-proxy-only
still an option |
docs/LOCAL_APPLIANCE_ARCHITECTURE.md:168:| T11 | Retention sweep for `output/`+`uploads/` ?
**sweep tool shipped** ([engine #299](https://github.com/Khelsutra/badminton-highlight-
indexer/pull/299)): dry-run-first; purges stale uploads >7d + orphan `.partial-` + regenerable
frames/staged/proxies; **reels/DB/CSVs durable**; `--json` plan ready for a scheduler. Automated
scheduling ? [engine #301](https://github.com/Khelsutra/badminton-highlight-indexer/issues/301).
| ? | owner ? (uploads 7d, keep reels ? 2026-06-21) | ? | [engine
#299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299) |
docs/LOCAL_APPLIANCE_ARCHITECTURE.md:178:**Blocking gates:** (1) **edge-auth** ? Caddy basicauth
(self-contained) vs cloudflared+Cloudflare Access (matches the two-host design, no inbound
ports)? (2) **retention defaults** ? **window DECIDED** (uploads **7d**, reels durable; [engine
#299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299) tool + scheduling
[#301](https://github.com/Khelsutra/badminton-highlight-indexer/issues/301)); **still open:**
auto-delete originals after compile vs keep proxies+intervals only (privacy/consent). (3) **per-
video Gemini/GPU cost ceiling** ? what cap, hard-fail vs warn? (4) **GPU host choice** ? DO GPU
droplet / GCE GPU VM / Cloud Run+GPU (and is a containerized worker worth building for Cloud
Run)? (5) **P5 bridge ownership** ? who builds the orchestration endpoint + how it coordinates
with the active worker sessions.
docs/LOCAL_APPLIANCE_ARCHITECTURE.md:188:- [ ] Retention sweep in place (**tool shipped**
[engine #299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299); **auto-
schedule** [engine #301](https://github.com/Khelsutra/badminton-highlight-indexer/issues/301));
`GEMINI_API_KEY` rotated; `allowed_video_root` set.
```

```
=== CF token / broad token references in khelsutra deploy ===
docs/ANNOTATE_AND_TRAIN.md:52:- **Marketing** `[verified]`: clone `site/public/capture.html`
skeleton + brand tokens (`index.html:25-32`); reuse `.honest`/`.notes` components (`:152-158`,
used `:351-354`, `:400-415`); add a `own-model` alias in `site/server.js:49-51` (Worker
`env.ASSETS.fetch` needs no change, `site/src/index.js:84`); cross-link nav `:218-223`, the
"gets sharper" section `:327-356`, footer `:528`; screenshots under `site/public/img/`
(precedent `site/og.png` + `make_og.py`).
docs/FRIENDLY_AND_MULTILINGUAL.md:183:- `2026-06-22` ? **A3
([#74](https://github.com/avidullu/khelsutra/pull/74)) landed ? the marketing `site/` recording-
education + marketing copy now render in kn + hi. **Translated the homepage "Record it right"
gist + the full 1-page capture checklist + the existing `site.` copy to **Kannada + Hindi**
(-40 new `site.record.`/`site.capture.` keys; en-authored, kn/hi translated, es/da/id en-
seeded ? key-parity green). The shim gains a **data-i18n-html** form so inline `<b>` emphasis
survives translation; **Noto Sans Kannada + Devanagari** bundled via Google Fonts on all 3 pages
(+ `--font` fallback) so kn/hi render without tofu. Translations **AI-drafted ? adversarially
verified** by a 6-agent workflow (meaning/fluency/integrity × kn/hi; 0 issues in the A3 keys)
and a 4-lens code review (folded its a11y findings ? switcher group `aria-label` now localized
via `lang.label`, default `aria-pressed` for the no-JS baseline ? and 3 test-coverage findings
incl. a byte-for-byte HTML?en-catalog drift test). **Web 173 + site 68 green**; real-browser
```

check (kn capture page renders Kannada, `**<b>**` preserved, no console errors). kn/hi remain ****native-review-pending**** (font glyphs = owner eyeball gate). The full-catalog audit also surfaced pre-existing ****B5/B6 SPA-console**** translation bugs (brand transliteration, "engine"/"server" jargon, a code-token leak, over-Sanskritized Hindi) ? flagged as a separate follow-up.

docs/FRIENDLY_AND_MULTILINGUAL.md:189:- `2026-06-22` ? ****own-model** body content localized to kn + hi ([#86](https://github.com/avidullu/khelsutra/pull/86)) ? closes engineering follow-up (3) above.****** The developer/power-user `own-model.html` was English-only on kn/hi (only its switcher + Request-access link were localized). Wired its ****full body**** ? hero, the annotate/train/setup steps, the moat grid, the honest-framing fences, "what this is not", CTA and footer ? through the shim (****52 new `site.ownModel.\*` keys****; `data-i18n` + `data-i18n-html` for inline `**<b>**/**<i>**/**<a>**/**<code>**/**<span>**`), keeping `**<pre>**`/CLI blocks, file paths, the `ending\_reason` enum, brand/tool tokens (KheISutra/VLC/rally-annotator/MIT/GPU/LLM) and the VLC UI button labels ****literal****, and every honest caveat intact (`**<title>**`/meta left English, consistent with A3). en authored ****byte-for-byte**** (jsdom-extracted ? the A3 drift guard validates it); es/da/id en-seeded; ****kn/hi AI-drafted ? 3-lens adversarial verify (accuracy/fluency/integrity) ? revise**** via a 10-agent workflow (16 kn + 10 hi findings folded), reusing the established kn/hi register; the shipped `requestAccess` translation preserved. New tests: own-model body-wiring assertion + kn/hi in-script render check + a ****tag-multiset/literal-token integrity guard****; the web `catalog-invariants` `**<code>**`-leak rule (#77) ****scoped**** to allow inline code only on `site.ownModel.\*`. ****Site 75 + web 215 green; web typecheck/coverage (93.4%, ? floor)/build green.**** Real-browser pass (kn + hi render, `**<b>**` preserved, code blocks literal, ****zero console errors****); screencast `khelsutra-ownmodel-i18n-desktop.webm` + per-locale stills in `khelsutra-screencast/`. kn/hi remain ****native-review-pending**** (the two human gates are unchanged). Built in an isolated `git worktree` off `origin/master` (the shared checkout had churned to `deploy/kit-live-verified` mid-session ? concurrency convention vindicated).

docs/LOCAL_APPLIANCE_ARCHITECTURE.md:137:| ****Provisioning-credential leak**** | a DO API token / cloud creds could leak if pasted in chat or committed (cf. the ****already-rotated Spaces key**** exposed in chat, `DECOUPLED\_COMPUTE/HANDOFF.md`) | authenticate the CLI ****on-box**** (`doctl auth init` interactive / `gcloud` already authed) so the secret never enters chat or the repo; least-privilege token, revoke after; creds 0600, never committed; ****creating paid GPU infra is a confirm-first action every time**** |

docs/LOCAL_APPLIANCE_ARCHITECTURE.md:173:****Owner priority (2026-06-20): T13 (GPU host) is elevated to run in parallel from the start**** ? owner provisions the host + token; the engine session lands native WASB; the UI MVP does ****not**** block on it (ships on Gemini).

docs/PER_RALLY_SELECTION.md:59:- ****QueryBar + `parseQuery()`**** ? a ****pure, deterministic, unit-tested* string ? `{filter, sort}` compiler over `duration` / `ordinal` / `shot\_count`**. Renders a ****parsed-pill**** ("filter: duration > 10s · sort: duration desc") so the interpretation is transparent and hand-correctable; greys out shot-type/player/score tokens with an inline "not available yet". No LLM, no network call.

docs/PER_RALLY_SELECTION.md:130:| P3 | `parseQuery()` + parsed-pill + example chips (unit-tested; greys out unavailable tokens) | P2 | No | ? | (this PR) |

docs/PER_RALLY_SELECTION.md:155:- [x] `parseQuery()` covers length/count/order/shot_count, greys out unavailable tokens, shows a visible parsed-pill, and is unit-tested. ***(P3)***

docs/PORTABILITY.md:40: via `git archive` (****private-repo-safe**** ? no clone, token, or secret on the VM), installs Node (?24,

28. assistant (2026-06-22T01:57:27.844Z)

#301 confirmed as a khelsutra-referenced engine issue for auto-scheduling the retention sweep. Now let me find the broad CF token issue specifically and the native-speaker review status in the khelsutra deploy docs. Let me read the live-verified deploy kit and the alpha deploy doc.

29. user (2026-06-22T01:57:29.259Z)

```
=== deploy dir contents ===
deploy/DEPLOY_ALPHA.md
deploy/cloudflared.config.example.yml
deploy/droplet/README.md
deploy/droplet/config.json.example
deploy/droplet/engine.env.example
deploy/droplet/khelsutra-engine.service
site/src/deploy-config.test.js
```

30. user (2026-06-22T01:57:33.492Z)

C:\Users\avidu\Projects\khelsutra\deploy\DEPLOY_ALPHA.md:102:3. ****Policy:**** Action ***Allow***, rule ***Emails*** ? your alpha testers' addresses (free ? 50 users ? fits 10?25 testers). Use one-time-PIN (email OTP) so testers need no Cloudflare account.

31. assistant (2026-06-22T01:57:37.616Z)

The CF token concern isn't documented in the deploy kit yet. The task brief flags it as something the owner should revoke ? this is the cloudflared tunnel setup, which by default issues a broadly-scoped token. Let me check the cloudflared config example and the droplet README to ground the token scope claim.

32. user (2026-06-22T01:57:38.142Z)

```
1      # cloudflared tunnel config ? KhelSutra Studio ENGINE (api.khelsutra.guru ? FastAPI on
the box).
2      #
3      # Copy to the engine host (e.g. /etc/cloudflared/config.yml or
~/cloudflared/config.yml), fill in
4      # <TUNNEL_UUID>, then run cloudflared as a service. The tunnel dials OUT to Cloudflare,
so NO inbound
5      # port is opened on the box ? the engine stays bound to 127.0.0.1:8000 (its safe
default), and the
6      # only way in is through Cloudflare Access. See DEPLOY_ALPHA.md.
7      #
8      # The SPA (app.khelsutra.guru) is NOT served here ? it ships on Cloudflare Pages
(DEPLOY_ALPHA.md §3).
9      # This tunnel carries ONLY the engine API.
10
11     tunnel: <TUNNEL_UUID>
12     credentials-file: /etc/cloudflared/<TUNNEL_UUID>.json
13
14     ingress:
15       # The engine FastAPI control plane (POST /api/process, GET /api/jobs, /api/query,
/api/compile,
16       # GET /api/highlights, /api/health). Bound to 127.0.0.1:8000 on the box ? never
0.0.0.0.
17       - hostname: api.khelsutra.guru
18         service: http://127.0.0.1:8000
19         originRequest:
20           # The async pipeline (fetch ? segment ? stitch) can run for minutes; don't let the
tunnel
21           # time the request out before the job is submitted (submit returns 202 fast, but
be generous).
22           connectTimeout: 30s
23           noTLSVerify: false
24
25       # A cloudflared ingress MUST end with a catch-all.
26       - service: http_status:404
27
```

33. assistant (2026-06-22T01:57:41.597Z)

Now let me check the DEPLOY_ALPHA.md for the tunnel-creation step (where a broad token may be minted) and the native-speaker review status. Let me read the relevant parts.

34. user (2026-06-22T01:57:42.243Z)

```
1      # Deploy KhelSutra Studio for alpha ? a shareable page, no SSH
2
3      > **Status:** `READY` (owner executes the cloud steps) . **Topology:** split ? SPA on
Cloudflare Pages + engine on an `api.` cloudflared tunnel, Cloudflare Access over both (the
engine's `HOSTING_PLAN.md` §2 alpha plan). . **Cost:** ~$0/mo standing (Pages + Tunnel + Access
```

are free ?50 users) + Gemini per-run (~\$0.05/match).

```
4
5     This turns the local-only flow (run engine + SPA on your box, drive it yourself) into a
6     **page you can share with alpha testers** ? they open `app.khelsutra.guru`, sign in (email
7     allowlist), point at a video, and get a reel. No SSH, no home IP exposed, no inbound ports.
8
9     ```
10    tester browser
11    ? https
12    ?
13    Cloudflare edge (free): DNS · TLS · Access (email-OTP allowlist) in front of BOTH
14    hostnames
15    ??? app.khelsutra.guru ? Cloudflare Pages          (the web/ SPA; built with
16    VITE_API_BASE)
17    ??? api.khelsutra.guru ? cloudflared Tunnel (outbound-only) ?? 127.0.0.1:8000 on
18    YOUR box
19
20    ? FastAPI engine
21    ? (Gemini
22    segmenter; GPU optional)
23    ? async jobs,
24    SQLite, output/ on disk
25    ```
26
27    - **No inbound ports.** `cloudflared` dials out; the engine stays bound to
28    `127.0.0.1:8000`. The box's home/public IP is never exposed.
29    - **Auth at the edge.** Cloudflare Access gates both hostnames; the engine *also*
30    verifies the `Cf-Access` JWT in-app (defence in depth ? `security.edge_auth_enabled`), so a
31    direct hit to the tunnel can't spoof identity.
32    - **Cross-origin, on purpose.** `app.` (SPA) and `api.` (engine) are different origins ?
33    the SPA is built with `VITE_API_BASE=https://api.khelsutra.guru/api` (PR #64) and the engine
34    locks CORS to the SPA origin via `RALLY_CORS_ALLOW_ORIGINS` (no engine code change ? it's
35    already env-driven).
36
37    > **Alternative (single-origin):** if you'd rather run **one box** that serves the SPA
38    *and* the API behind one origin (Caddy reverse-proxy, no Pages, no cross-origin), see
39    `LOCAL_APPLIANCE_ARCHITECTURE.md` D1. This runbook ships the **split** topology you chose
40    (stable Pages-hosted page + a thin API tunnel).
41
42    ---
43
44    ## 1. Prerequisites
45
46    - A **box** to run the engine: the CPU droplet `168.144.126.176` (Gemini-only ? no GPU;
47    fine for alpha) **or** your Windows+WSL GPU box (real WASB). `ffmpeg` + Python 3.8+ installed;
48    the `badminton-highlight-indexer` engine checked out.
49    - A **Cloudflare account** with the `khelsutra.guru` zone (you already run the marketing
50    site there).
51    - `cloudflared` installed on the box (`https://developers.cloudflare.com/cloudflare-
52    one/connections/connect-networks/downloads/`).
53    - A **fresh** `GEMINI_API_KEY` (rotate the chat-shared one ? it uploads downsized chunks
54    to Google; this is a paid + cloud action).
55
56    ---
57
58    ## 2. Engine (the `api.` side)
59
60    ### 2.1 Install (with the auth extra)
61
62    ```bash
63    cd badminton-highlight-indexer
64    pip install -e .[auth]          # PyJWT[crypto] ? REQUIRED when edge auth is on, else
65    every authenticated request is rejected 401
66    ```
```

```

46     ### 2.2 The safe-exposed config
47     Create / edit `config.json` on the box. **Where it must live:** the engine reads
`$RALLY_APP_HOME/config.json` if `RALLY_APP_HOME` is set, else `./config.json` relative to *the
directory you launch the server from * ? **not** necessarily the repo root. If the engine reads a
different (or absent) file it silently falls back to all-defaults, which means **edge auth is
OFF**. To make writer and reader agree by construction, write it with `python -m backend.main
--setup` (it targets the exact path the server reads) and edit *that* file:
48     ```jsonc
49     {
50         "security": {
51             "edge_auth_enabled": true,                                // turn the CF-Access JWT verify ON
52             "allowed_video_root": "/srv/khelsutra/incoming", // the path-jail ? REQUIRED when
exposed
53             "upload_dir": "/srv/khelsutra/incoming/uploads", // MUST sit UNDER
allowed_video_root (see §7)
54             "cf_team_domain": "https://<your-team>.cloudflareaccess.com",
55             "cf_access_aud": "<the CF Access application AUD>" // filled in §4
56         },
57         "indexing": {
58             "default_segmenter": "gemini" // ? CPU box MUST set this ? the shipped default
`yolo_hybrid` needs torch/GPU, and the SPA sends no segmenter, so every job fails "No module
named 'torch'" without it
59         }
60     }
61     ```
62     And export the env (a `.env` or your shell):
63     ```bash
64     export GEMINI_API_KEY="<your-rotated-key>"
65     export RALLY_CORS_ALLOW_ORIGINS="https://app.khelsutra.guru" # lock CORS to the SPA
origin
66     export RALLY_ALLOWED_HOSTS="api.khelsutra.guru" # REQUIRED ? see ? below;
without it EVERY tunneled request 400s
67     export RALLY_BIND_HOST="127.0.0.1" # keep the loopback bind
(ccloudflared dials 127.0.0.1:8000)
68     ```
69
70     > **? Two env vars you MUST set, or the tunnel is dead-on-arrival.** With
`edge_auth_enabled: true`, the engine (a) flips its bind to `0.0.0.0` and (b) keeps the DNS-
rebinding **Host-header guard** at its loopback-only default ? so it boots "healthy" but then
**400's every request `cloudflared` forwards with `Host: api.khelsutra.guru`** (including the
§6 `/api/health` check). The boot only *warns* about this (`EXPOSED_HOST_ALLOWLIST_LOOPBACK`),
it does not stop. Set **`RALLY_ALLOWED_HOSTS="api.khelsutra.guru"`** (the load-bearing fix ? the
Host allowlist is independent of the bind) and **`RALLY_BIND_HOST="127.0.0.1"`** (so the bind
stays loopback as §0/§6 claim; `cloudflared` dials loopback anyway). *Verified against engine
`origin/master`: with the allowlist unset, `GET /api/health` with `Host: api.khelsutra.guru` ?
`400`; with it set ? `200`.*
71
72     > **The engine refuses to start unsafe.** The boot posture (engine PR #294) **fails
fast** if `edge_auth_enabled` is on but the `allowed_video_root` jail or
`cf_team_domain`/`cf_access_aud` is missing, and **warns** if `PyJWT` isn't installed. So a
half-configured exposure can't silently come up ? fix what it prints and restart.
73
74     ### 2.3 Run
75     ```bash
76     python -m backend.main --server # binds 127.0.0.1:8000 ? kept loopback by
RALLY_BIND_HOST (§2.2); with edge_auth ON and that var unset it would bind 0.0.0.0 instead
77     ```
78     Keep it alive with a process manager (`systemd` on the droplet, mirroring
`site/scripts/provision-vm.sh`'s unit; or `pm2`/`nssm` on Windows).
79
80     ---
81
82     ## 3. The SPA on Cloudflare Pages (the `app.` side)
83
84     The SPA ships as a **static Pages site**, git-connected like the marketing site (auto-

```

```

deploy on merge to `master`).
85
86 1. **Cloudflare dashboard ? Workers & Pages ? Create ? Pages ? Connect to Git** ? pick
the `khelsutra` repo.
87 2. **Build settings:**
88 - **Root directory:** `web`
89 - **Build command:** `npm run build`
90 - **Build output directory:** `web/dist`
91 - **Environment variable:** `VITE_API_BASE = https://api.khelsutra.guru/api` ? this
is what points the SPA at the engine tunnel (PR #64).
92 3. **Custom domain:** add `app.khelsutra.guru` to the Pages project.
93
94 > Every push to `master` now rebuilds + redeploys the SPA. To preview a change first,
Pages builds a per-PR preview URL.
95
96 ---
97
98 ## 4. Cloudflare Access (gate BOTH hostnames)
99
100 1. **Zero Trust ? Access ? Applications ? Add ? Self-hosted.**
101 2. **Application domain:** add **both** `app.khelsutra.guru` **and**
`api.khelsutra.guru` to the *same* application (one app ? one shared session cookie, so a tester
signs in once and both the SPA and its API calls are authorized).
102 3. **Policy:** Action *Allow*, rule *Emails* ? your alpha testers' addresses (free ? 50
users ? fits 10?25 testers). Use one-time-PIN (email OTP) so testers need no Cloudflare account.
103 4. **Copy the application `AUD` tag** ? paste into the engine's `config.json`
`security.cf_access_aud` ($2.2), and set `cf_team_domain` to `https://<your-
team>.cloudflareaccess.com`. Restart the engine.
104 5. **CORS preflight ?:** the browser sends an unauthenticated `OPTIONS` preflight to
`api.` for the cross-origin POSTs. In the Access application's **Settings ? enable ?Bypass
options requests to CORS preflight?** (or add an explicit OPTIONS bypass), else the preflight is
redirected to the login page and every API call fails. The engine's CORS middleware
(`allow_credentials=false`, no cookies) then answers the preflight.
105
106 ---
107
108 ## 5. The tunnel (engine ? Cloudflare)
109
110 ```bash
111 cloudflared tunnel login # browser auth to your zone
112 cloudflared tunnel create khelsutra-studio # note the <TUNNEL_UUID>
113 # copy deploy/cloudflared.config.example.yml ? /etc/cloudflared/config.yml, fill
<TUNNEL_UUID>
114 cloudflared tunnel route dns khelsutra-studio api.khelsutra.guru
115 cloudflared tunnel run khelsutra-studio # or install as a service:
`cloudflared service install`
116 ```
117 See `deploy/cloudflared.config.example.yml` for the ingress (only `api.khelsutra.guru` ?
http://127.0.0.1:8000`).
118
119 ---
120
121 ## 6. Verify (the posture test + the CUJ)
122
123 **Exposure posture (must hold ? this is what makes it safe to share):**
124 - From OUTSIDE the box: `curl --max-time 5 http://<box-public-ip>:8000/api/health` ?
**refused / timeout** (the engine is `127.0.0.1`-bound; no inbound port). ? The box is not
directly reachable.
125 - `curl -i https://api.khelsutra.guru/api/health` **without** a CF session ? a **302 to
the Access login** (not a 200). ? The gate is in front.
126 - Open `https://app.khelsutra.guru` in a browser ? CF Access login ? after sign-in, the
SPA loads and `/api/health` returns 200 (same session cookie). ?
127
128 **The shareable CUJ (what a tester does):** open `app.khelsutra.guru` ? sign in ?
**upload a video from their device** (the SPA chunked-upload lands it in `upload_dir`, inside

```



```

the jail) ? **Process** ? watch progress ? pick rallies ? **Build** ? **download the reel**. All
from the page. *(Pasting a `gs://` URI does NOT work on this box ? see §7: the
`allowed_video_root` jail rejects every non-local URI. Browser upload is the alpha path.)*
129
130 ---
131
132 ## 7. Honest caveats / deferred
133
134 - **`upload_dir` must sit UNDER `allowed_video_root`.** If you enable browser upload, an
upload landing outside the jail is rejected by `/api/process` (a functional break, not a hole).
Keep `upload_dir` inside the jail (§2.2).
135 - **`gs://` / `s3://` / `gdrive://` URIs are REJECTED on the exposed box.** Setting
`allowed_video_root` (required to expose, §2.2) path-jails `/api/process` to *local* paths under
that root; the engine resolves a `gs://` URI to a local path that escapes the jail ? `400`
(pinned by engine `tests/test_api_endpoints.py::test_process_hosted_mode_rejects_nonlocal_uri`).
Cloud-URI ingest works **only** when `allowed_video_root` is unset (the local single-user
posture ? the opposite of this runbook). For alpha, the ingest path is **browser upload** (it
lands inside the jail). *(Scheme-exempting recognized buckets from the jail would be an engine
change, not a config flip.)*
136 - **You MUST set `edge_auth_enabled=true` to expose.** The boot posture (and the in-app
JWT verify) only engage then. Do not put the engine behind the tunnel with edge auth off.
137 - **Cross-origin reel download:** the reel streams from `api.` with `Content-
Disposition: attachment`, so it downloads even cross-origin ? but verify the filename in your
browser during the CUJ (cross-origin `` filename hints are ignored; the header is
the authority).
138 - **Upload size:** Cloudflare's free proxy caps request bodies at **100 MB** ? large
matches upload as ~95 MB chunks (the SPA chunked-upload path honours the engine's `max_part_mb`,
default **95** ; lower `security.max_part_mb` to shrink them). `gs://` bucket pre-staging is
**not** an option on this box (see the jail caveat above).
139 - **Box offline = engine offline.** The Pages SPA stays up and shows a friendly "engine
offline" banner (`/api/health` ping); processing resumes when the box is back. Beta removes this
with a queue (`HOSTING_PLAN.md` §3).
140 - **Gemini = paid + uploads footage to Google.** Keep the key rotated; use it only for
clips you're OK sending.
141
142 ---
143
144 ## 8. Teardown
145 `cloudflared tunnel cleanup khelsutra-studio` + delete the tunnel; remove the Pages
custom domain; disable the Access application; stop the engine service. The box's `output/` +
`uploads/` retain originals ? sweep them per your retention policy before any wider exposure.
146
147 ---
148 ### Changelog
149 - 2026-06-22 ? **correctness pass** (adversarial verify against engine `origin/master` +
a hermetic repro). Fixed two would-break-the-deploy gaps: (1) **`RALLY_ALLOWED_HOSTS` was
missing** ? with `edge_auth` on, the Host-header guard 400s every tunneled request until it's
set (added it + `RALLY_BIND_HOST` to §2.2; repro confirmed 400?200); (2) **`gs://` ingest is
rejected** by the `allowed_video_root` jail on the exposed box (dropped the `gs://` promise from
§6/§7; alpha path = browser upload). Minor: 401-not-500 wording (§2.1), `config.json`
location/`--setup` note (§2.2), bind comment (§2.3), ~95 MB chunk size (§7).
150 - 2026-06-21 ? created. Split-topology alpha deploy kit (cloudflared + Pages + CF
Access) consuming the engine exposure-safety boot posture (engine #294) + the SPA
`VITE_API_BASE` (#64). Pairs with `LOCAL_APPLIANCE_ARCHITECTURE.md` (single-origin alternative)
+ the engine `HOSTING_PLAN.md` §2.
151

```

35. assistant (2026-06-22T01:57:48.633Z)

Now I have the full picture on the CF token. The DEPLOY_ALPHA §5 uses `cloudflared tunnel login`, which mints a **cert.pem with account-wide scope** (zone edit + tunnel create for the whole account) ? that broad credential lives on the box and should be revoked/scoped down after the tunnel is created. That's the "broad CF token the owner should revoke" the task flags. Confirmed via the runbook: §8 teardown cleans the tunnel but doesn't mention revoking the broad login cert.

Now let me confirm the native-speaker review status (it's the launch gate) and check whether Finding B has any existing tracking issue anywhere. Let me also verify there's no engine-side issue list I'm missing.

36. user (2026-06-22T01:57:50.866Z)

```
=== native review status in FRIENDLY_AND_MULTILINGUAL ===
docs/FRIENDLY_AND_MULTILINGUAL.md:119:- **End-to-end (manual, owner machine ? per I18N_PLAN
$6):** set the real OS/browser to Kannada, then Hindi ? load both surfaces ? confirm the right
language renders **and the Noto fonts display** (the cloud CI container can't render fonts; the
owner eyeball is the final gate).
docs/FRIENDLY_AND_MULTILINGUAL.md:138:| B5 | **Kannada (`kn`) catalog** ? render ? *SPA done
(AI-draft, native review pending); `site/` rides B3* | B | B2/B4 | D-LOCALES ? | ? |
[#47](https://github.com/avidullu/khelsutra/pull/47) |
docs/FRIENDLY_AND_MULTILINGUAL.md:139:| A3 | Localize **recording-education** (capture +
homepage gist) + the `site.*` marketing copy to **kn + hi** (Noto on the site; AI-draft,
adversarially verified, native review pending) | A | B3/B5 | D-LOCALES ? + D-FONT ? (site:
Google-Fonts Noto) | ? | [#74](https://github.com/avidullu/khelsutra/pull/74) |
docs/FRIENDLY_AND_MULTILINGUAL.md:142:| B6 | **Hindi (`hi`) catalog** ? render (the
extensibility proof ? just a catalog) ? *SPA done (AI-draft, native review pending)* | B | B5 |
D-LOCALES ? | ? | [#48](https://github.com/avidullu/khelsutra/pull/48) |
docs/FRIENDLY_AND_MULTILINGUAL.md:147:**? MILESTONE ? SPA fully trilingual (2026-06-21):** the
`web/` Studio SPA now speaks **en + kn + hi** end to end (B1?B2?B4?B5?B6, all merged). A
**Kannada-default browser auto-renders Kannada**, and the switcher walks kn ? hi ? en ? **proven
by a Playwright screencast** with `locale: kn-IN` (`khelsutra-i18n-screencast.mp4`; frames
captured per locale + `` tracked). This meets the owner's success criterion **on the
SPA surface**. **B3 has since landed** ? the `site/` marketing surface now carries the shared
`t()/`data-i18n` shim, a persistent en/kn/hi switcher, and matching negotiation (client shim +
`server.js`; $6.5 **N8** green), with **en-only marketing content** for now (the
shim/switcher/negotiation are the proof; kn/hi *marketing* translation rides **A3**). **A3 has
since landed** ? the `site/` recording-education (homepage gist + capture checklist) and the
`site.*` marketing copy now render in **kn + hi** (Noto bundled on the site), so a Kannada/Hindi
browser auto-gets localized recording instructions. **A6** (plain-language access form,
[#75](https://github.com/avidullu/khelsutra/pull/75)) and **A7** (camera-placement diagram,
[#76](https://github.com/avidullu/khelsutra/pull/76)) have since landed, and the **kn/hi SPA-
console native-review pass** shipped ([#77](https://github.com/avidullu/khelsutra/pull/77)).
**Still open for the full $9 DoD:** a **native-speaker review** of the *remaining* AI-drafted
catalogs (es/da/id + the reel-watermark strings; the kn/hi SPA console is done via #77) before a
wide public launch. (**A2** WhatsApp/social share is **deferred out of v1 scope** ? not a DoD
blocker.) Do **not** archive until both surfaces render kn/hi *content*. ? (both now do, pending
the remaining native review).
docs/FRIENDLY_AND_MULTILINGUAL.md:159:- [ ] **Manual cross-check (owner machine):** OS/browser
set to Kannada and to Hindi each render the respective language on BOTH surfaces, with Noto
fonts displaying. *(Owner eyeball ? pending; cloud CI can't render font glyphs. Closest proxy:
the Playwright screencasts in `khelsutra-screencast/` ? `khelsutra-site-i18n-desktop.mp4` (site,
kn) + `khelsutra-i18n-screencast.mp4` (SPA, `locale: kn-IN`)).*
docs/FRIENDLY_AND_MULTILINGUAL.md:161:- [ ] **Noto Sans Kannada + Devanagari render correctly**
? owner eyeball on a real machine. *(SPA self-hosts OFL Noto (B4); the site loads Noto via
Google Fonts (A3). Pending the owner's device check.)*
docs/FRIENDLY_AND_MULTILINGUAL.md:180:- `2026-06-21` ? **A1 (#60) + A4 (#61) landed.** A1 = the
plain-language `en` copy pass (de-jargon). A4 = the non-tech / phone-first pass: 244px tap
targets across the SPA (Tailwind `min-h-11`/`h-11`), enlarged parsed-pill + sample chips, and
**hover-only `title=` text surfaced as visible DOM text** (the QueryBar "not available yet"
reason + per-card shot-count confidence) ? a phone never shows `title=`. 128 web tests green;
reviewed via a 3-lens adversarial workflow. **Still open in workstream A:** A5 (friendly engine-
error map on the build path), A8 (background-processing reassurance / status chips), A9 (tap-
only filters), A3 (recording-education kn/hi), B3 (site/ trilingual) + native review.
docs/FRIENDLY_AND_MULTILINGUAL.md:181:- `2026-06-21` ? **? SPA MILESTONE: fully trilingual
(owner directive "take it to completion + screencast").** B2 (#45, full string extraction + key-
parity/no-bare-strings CI) ? B4 (#46, self-host OFL Noto) ? B5 (#47, Kannada) ? B6 (#48, Hindi ?
the extensibility proof) all merged. 99 web tests green incl. N1/N2 browser-detection + B5/B6
render + key-parity. **Playwright screencast** (`locale: kn-IN`): a Kannada-default browser
auto-renders Kannada ? switch hi ? switch en, each with `` tracked
(`khelsutra-i18n-screencast.mp4` + per-locale frames). kn/hi are **AI-drafted ? native review
pending**. **Remaining for full $9 DoD:** B3 (the `site/` marketing surface), A3 + the UX
```

A-items, native-speaker translation review.

docs/FRIENDLY_AND_MULTILINGUAL.md:183:- `2026-06-22` ? **A3

([#74](https://github.com/avidullu/khelsutra/pull/74)) landed ? the marketing `site/` recording-education + marketing copy now render in kn + hi.** Translated the homepage "Record it right" gist + the full 1-page capture checklist + the existing `site.\*` copy to **Kannada + Hindi** (-40 new `site.record.\*`/`site.capture.\*` keys; en-authored, kn/hi translated, es/da/id en-seeded ? key-parity green). The shim gains a **`data-i18n-html`** form so inline `**` emphasis survives translation; **Noto Sans Kannada + Devanagari** bundled via Google Fonts on all 3 pages (+ `--font` fallback) so kn/hi render without tofu. Translations **AI-drafted ? adversarially verified** by a 6-agent workflow (meaning/fluency/integrity × kn/hi; 0 issues in the A3 keys) and a 4-lens code review (folded its ally findings ? switcher group `aria-label` now localized via `lang.label`, default `aria-pressed` for the no-JS baseline ? and 3 test-coverage findings incl. a byte-for-byte HTML?en-catalog drift test). **Web 173 + site 68 green**; real-browser check (kn capture page renders Kannada, `**` preserved, no console errors). kn/hi remain **native-review-pending** (font glyphs = owner eyeball gate). The full-catalog audit also surfaced pre-existing **B5/B6 SPA-console** translation bugs (brand transliteration, "engine"/"server" jargon, a code-token leak, over-Sanskritized Hindi) ? flagged as a separate follow-up.****

docs/FRIENDLY_AND_MULTILINGUAL.md:185:- `2026-06-22` ? **A7

([#76](https://github.com/avidullu/khelsutra/pull/76)) landed ? camera-placement diagram on the capture page.** Added an **accessible inline SVG** (`role="img" + localized `` side-view showing the load-bearing placement cues a coach can't get from text alone: the **elevated camera behind the baseline**, the **~30° downward tilt**, and the **full-court field-of-view cone** (players within frame). A numbered ??? legend (HTML, so it wraps) maps the markers to the rules and is localized **en/kn/hi** (`site.capture.diagram.\*`, reusing the A3 capture terms) ? consistent with the now-trilingual capture page. The text "Place the camera" step remains (the diagram complements it). Tests: a diagram-structure + accessible-`<title>` + localized-legend assertion; kn/hi in-script check incl. `diagram.camera`; the A3 drift test auto-covers the new keys; site 70 + web 173 green. **Workstream A is now A1/A3/A4/A5/A6/A7/A8/A9 ?** (remaining: A2 WhatsApp share; the B5/B6 SPA-console native-review pass; full native review).</p></div><div data-bbox="64 458 825 750" data-label="Text"><p>docs/FRIENDLY_AND_MULTILINGUAL.md:189:- `2026-06-22` ? **own-model body content localized to kn + hi ([#86](https://github.com/avidullu/khelsutra/pull/86)) ? closes engineering follow-up (3) above.** The developer/power-user `own-model.html` was English-only on kn/hi (only its switcher + Request-access link were localized). Wired its **full body** ? hero, the annotate/train/setup steps, the moat grid, the honest-framing fences, "what this is not", CTA and footer ? through the shim (**52 new `site.ownModel.\*` keys**; `data-i18n` + `data-i18n-html` for inline `<b><i>/<a>/<code>/<span>`), keeping `<pre>/CLI blocks, file paths, the `ending_reason` enum, brand/tool tokens (KhelSutra/VLC/rally-annotator/MIT/GPU/LLM) and the VLC UI button labels \*\*literal\*\*, and every honest caveat intact (`<title>/meta left English, consistent with A3). en authored **byte-for-byte** (jsdom-extracted ? the A3 drift guard validates it); es/da/id en-seeded; **kn/hi AI-drafted ? 3-lens adversarial verify (accuracy/fluency/integrity) ? revise** via a 10-agent workflow (16 kn + 10 hi findings folded), reusing the established kn/hi register; the shipped `requestAccess` translation preserved. New tests: own-model body-wiring assertion + kn/hi in-script render check + a **tag-multiset/literal-token integrity guard**; the web `catalog-invariants` `<code>`-leak rule (#77) **scoped** to allow inline code only on `site.ownModel.\*`. **Site 75 + web 215 green; web typecheck/coverage (93.4%, ? floor)/build green.** Real-browser pass (kn + hi render, `<b>` preserved, code blocks literal, **zero console errors**); screencast `khelsutra-ownmodel-i18n-desktop.webm` + per-locale stills in `khelsutra-screencast/`. kn/hi remain **native-review-pending** (the two human gates are unchanged). Built in an isolated `git worktree` off `origin/master` (the shared checkout had churned to `deploy/kit-live-verified` mid-session ? concurrency convention vindicated).</p></div><div data-bbox="64 765 352 779" data-label="Section-Header"><h3>37. assistant (2026-06-22T01:58:00.463Z)</h3></div><div data-bbox="64 793 815 848" data-label="Text"><p>I now have everything grounded. Native-speaker review is confirmed open (launch gate, line 147: "Still open for the full \$9 DoD: a native-speaker review of the remaining AI-drafted catalogs (es/da/id + reel-watermark strings)"). The kn/hi SPA console review shipped via #77; the owner-eyeball font check + remaining-locale native review are still pending wide-launch gates.</p></div><div data-bbox="64 862 768 904" data-label="Text"><p>Let me do one final verification on Finding B's home: confirm there's no existing engine doc/issue tracking the "0-rally success masks a hard error" gap, and that BACKLOG doesn't already list it.</p></div><div data-bbox="64 918 320 934" data-label="Section-Header"><h3>38. user (2026-06-22T01:58:03.495Z)</h3></div>

```
=== BACKLOG: any mention of job-status / 0-rally / observability ===
85:## Observability reports ? shipped; intersections with the low-regret items (2026-06-07)
86:*(source: per-video observability feature ? indexer PR #41/#42/#43, collector PR #11/#12;
shared lib `obsreport`)*
89:- ? **Per-video observability reports shipped in BOTH tools** (a TECHNICAL report + a
CUSTOMER-friendly summary), via the **soft-dependency** lib `obsreport`
(github.com/avidullu/sports-obsreport, MIT, v0.1.3). Each run writes `
```

Backlog ? deferred items to revisit

Single tracker for work we consciously deferred, so nothing gets lost between sessions. Each item records **what**, **why** it's deferred, and **the trigger to revisit**. Every entry traces to a real decision/PR/doc (no speculative work).

> Maintained as a version-controlled doc for now. Say the word and I'll mirror any/all of these to **GitHub issues** instead (or in addition).
> Status legend: ? blocks something · ? do when triggered · ? nice-to-have/cleanup.

Last updated: 2026-06-10.

Owned-model / multisport (source: `OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md`, `NEXT\_STEPS.md`)

- ? **Rename the repo off "badminton"** ? owner-emphatic "ASAP" action; the project is multisport. Touches remote name + README/CLAUDE.md framing + any hardcoded URLs. **Revisit:** owner action, before broad external sharing.
- ? **Quantify cost-to-Gen-2 (the largest open risk)** ? benchmark real VRAM/token/\$ for the VLM SFT+GRPO on a rented A100/L40S (ms-swift multimodal-GRPO examples use 678 GPUs) **before** standing up any owned serving stack. **Revisit:** before Phase-2 commitment.
- ? **Define the ship-gate F1@tIoU target** ? "very high P/R" must be operationalized (floor = beat heuristic LOVO 0.480; learned is now 0.626 post metric-fix). **Revisit:** at M0.4 harness.

Engineering hardening deferred from the 2026-06-10 audit sweep

- ? **App-factory / lifespan + lazy torch-cv2 imports** ? `uvicorn backend.main:app` leaves globals None, and importing the API pulls the GPU stack. Belongs with the async-job slice (#16). **Revisit:** when the async slice starts.
- ? **WSL temp/frame/cache + Gemini remote-file GC** ? staged videos, PNG frame dumps, `\_wasbcache`, and orphaned Gemini uploads are never cleaned. Needs a retention policy. **Revisit:** before any volume/batch processing.
- ? **shlex-quote WSL config values** ? **DONE 2026-06-11 (#97):** ``detectors/base.py::wsl_tilde_quote`` quotes everything after a leading `~/`, so quoting and tilde expansion coexist; applied to every `~`-capable arg in both runners; ``tests/test_wsl_tilde_quoting.py`` pins the command shapes (no quoted tilde can ever reach `\_wsl\_bash`). Found live: the naive quoting had silently broken wasb_hybrid staging for the default `~/clips`. Residual for tenancy-era: quote the remaining trusted-config interpolations (repo_dir/conda_env) if config ever becomes per-tenant.

Monetization / licensing

(source: `archives/MONETIZATION\_AUDIT.md` §7, ADR-005)

- ? **Ultralytics Enterprise pricing** ? get a quote **only if** the torchvision/ByteTrack detector (PR #8) proves insufficient on real footage. Currently **not needed** (swap done). Quote-only pricing. **Revisit:** if detector quality forces reconsidering YOLOv8.
- ? **Codec patent thresholds (AVC/HEVC/AAC)** ? confirm exact Via-LA (AVC) / Access Advance (HEVC) royalty rates & free thresholds **with legal counsel** before relying on the "small SaaS is under threshold" reading. Mitigation already chosen: emit royalty-free AV1/Opus output. **Revisit:** before commercial launch.
- ? **Per-checkpoint weight/dataset licenses** ? if we borrow pretrained weights (YOLOX, ActionFormer, WASB, TrackNetV3), the **code** license ? the distributed **weights/dataset** terms. Either train our own or confirm each checkpoint. **Revisit:** when adopting any borrowed

weights.

- ? **Real GPU \$/video & latency** ? benchmark the three LLM-step options (offline ActionFormer vs self-hosted Qwen2.5-VL-7B vs Gemini paid-tier) on the target VM before pricing the product.
Revisit: when choosing the production LLM step.

Detector swap follow-ups

(source: PR #8, ADR-005)

- ? **supervision` ByteTrack deprecation** ? `sv.ByteTrack` is deprecated as of supervision 0.28.0 and **removed in 0.30.0**; `requirements.txt` is pinned `<0.30.0` as a stopgap. Migrate to `ByteTracker` from the separate `trackers` package before lifting the cap. **Revisit:** before bumping supervision past 0.30.

- ? **Real-video validation of the swap** ? the torchvision + ByteTrack Stage-1 is unit-tested with mocks but **not yet validated on actual footage** for detection/tracking quality. (ADR-004 separately found player pixel-motion is a weak rally proxy on broadcast video, so expect tuning.) **Revisit:** when a real test clip + the pipeline are run together.

- ? **yolo_hybrid` naming** ? the registry key, `yolo_` config keys, and `yolo-hybrid-` DB detector tags keep the "yolo" name for back-compat despite no YOLO remaining. Optional rename. **Revisit:** if/when a clean break is worth the migration.

Golden-set subproject

(source: `docs/GOLDEN\_SET\_IMPLEMENTATION.md`, `docs/GOLDEN\_SET\_VENDORING.md`)

39. assistant (2026-06-22T01:58:17.162Z)

Confirmed: BACKLOG (last updated 2026-06-10) has NO entry for the "0-rally success masks a hard error" gap nor any post-deploy observability item. The closest is the `obsreport` per-video reports (a different layer ? analysis reports next to the video, not job-status). Finding B is genuinely untracked.

I have all the grounding I need. Let me compile the prioritized follow-ups.

Summary of grounded findings:

- **Finding B (untracked, real):** `backend/pipeline/segmenters/gemini.py:92-99,514-516` ? missing `GEMINI\_API\_KEY` ? `\_resolve\_provider` logs error + returns `None` ? `process\_video` returns `[]`. `backend/api/job\_worker.py:76-90` records `status='done'` for any non-raising segmenter result, so a 0-rally "success" hides the hard config error. Asymmetry: the eval path `gemini\_refine.py` raises SystemExit on missing key (CODE_MAP:179), but the live segmenter swallows it. Precedent #271 added a LOUD log warning for the *WASB worker* 0-segment case ? but only in logs, not job status, and not the Gemini path. Not in BACKLOG (2026-06-10), DOC_STATUS, or NEXT_STEPS.

- **Retention scheduling:** engine #299 shipped the sweep TOOL (dry-run default, `--json` for a cron wrapper); auto-scheduling = engine #301 (open), referenced in khelsutra `LOCAL\_APPLIANCE\_ARCHITECTURE.md`. Now load-bearing because the alpha box accepts real uploads with a 7d privacy window.

- **CF broad token:** DEPLOY_ALPHA \$5 uses `cloudflared tunnel login` ? account-wide cert.pem on the box; \$8 teardown doesn't cover revoking it.

- **Native-speaker review:** open launch gate (FRIENDLY_AND_MULTILINGUAL \$9 DoD line 147) for es/da/id + reel-watermark strings + owner font eyeball; kn/hi SPA console done via #77.

40. user (2026-06-22T01:59:00.715Z)

Output does not match required schema: root: must have required property 'next_items', root: must have required property 'stale_doc_corrections'

41. assistant (2026-06-22T01:59:04.023Z)

The schema requires `next\_items` and `stale\_doc\_corrections` as top-level properties. I had a stray `` wrapper. Let me resubmit with the correct structure.

42. user (2026-06-22T01:59:32.673Z)

Output does not match required schema: root: must have required property 'next_items', root:

must have required property 'stale_doc_corrections'

43. assistant (2026-06-22T01:59:36.118Z)

The harness is rejecting JSON-as-string for the array/list parameters. Let me pass `next\_items` and `stale\_doc\_corrections` as native arrays rather than stringified JSON.

44. user (2026-06-22T02:00:05.203Z)

Output does not match required schema: root: must have required property 'next_items', root: must have required property 'stale_doc_corrections'

45. assistant (2026-06-22T02:00:09.404Z)

The XML-style array encoding for `next\_items` objects and `stale\_doc\_corrections` strings isn't being parsed into the required arrays. The validator reports both as missing, suggesting the nested array encoding failed. Let me try minimal valid arrays first with simple string items to confirm the encoding, then build up. Let me test the simplest case for the string array.

46. user (2026-06-22T02:00:10.503Z)

Structured output provided successfully

47. assistant (2026-06-22T02:00:13.988Z)

The JSON-string encoding works ? my second attempt must have had a malformed character in the long strings. Now let me submit the full payload using the same JSON-string array encoding that just succeeded.

48. user (2026-06-22T02:00:42.439Z)

Output does not match required schema: root: must have required property 'next_items', root: must have required property 'stale_doc_corrections'

49. assistant (2026-06-22T02:00:57.938Z)

Interesting ? the minimal version with short strings succeeded, but the full version fails. The difference must be a character in my long strings breaking the JSON parse (likely the `[ ]` literal, or `

50. user (2026-06-22T02:01:26.213Z)

Output does not match required schema: root: must have required property 'next_items', root: must have required property 'stale_doc_corrections'

51. assistant (2026-06-22T02:01:37.852Z)

Still failing. The minimal version succeeded, so the breakage scales with content. The most likely remaining JSON-breaker is the forward slash in paths ? no, JSON allows `/.`. Let me bisect: try the full `current\_status` and `next\_items` but with the proven-good minimal arrays for the two list fields, to isolate whether the long `current\_status` string is the problem.

52. user (2026-06-22T02:01:40.514Z)

Output does not match required schema: root: must have required property 'next_items', root: must have required property 'stale_doc_corrections'

53. assistant (2026-06-22T02:01:49.636Z)

The medium `current\_status` broke it while the short one passed. The breaker is in `current\_status` ? likely the `#` followed by digits, or `( #294 )`, or the `;` / parenthesis

combos. Most probably the parser is choking on the `#` character sequences. Let me test by replacing `#` with `PR` and removing parentheses-with-numbers in current_status.

54. user (2026-06-22T02:01:51.684Z)

Output does not match required schema: root: must have required property 'next_items', root: must have required property 'stale_doc_corrections'

55. assistant (2026-06-22T02:01:56.252Z)

Even with no special characters, the medium-length `current\_status` fails. So the breaker is LENGTH/total payload size, not characters. The minimal one-word version passed. There must be a total-output-size limit on this tool. Let me trim everything aggressively ? short `current\_status`, terse `why` fields ? to fit under the limit while keeping the substance.

56. user (2026-06-22T02:02:18.928Z)

Output does not match required schema: root: must have required property 'next_items', root: must have required property 'stale_doc_corrections'

57. assistant (2026-06-22T02:02:22.654Z)

Still failing despite trimming. The minimal test passed but this still fails. Let me bisect precisely: keep the full `current\_status` and minimal arrays, but this time I suspect the breaker is a specific token. Let me test the full `current\_status` from this last attempt with minimal arrays.

58. user (2026-06-22T02:02:24.702Z)

Output does not match required schema: root: must have required property 'next_items', root: must have required property 'stale_doc_corrections'

59. assistant (2026-06-22T02:02:29.263Z)

That's the key isolation. This `current\_status` is short but FAILS, while my earlier `test2` (similar length) also failed but the very first `test` ("test") passed. The breaker is the `[]` empty-bracket literal inside `current\_status` ? `returns [] :514-516`. The `[` likely confuses the harness's argument parser into thinking the field is an array. Let me confirm by removing the `[]`.

60. user (2026-06-22T02:02:30.981Z)

Structured output provided successfully

61. assistant (2026-06-22T02:02:34.498Z)

Confirmed ? the `[]` literal in `current\_status` was the breaker (the harness parsed `[` as an array start). I'll spell out "empty list" everywhere and avoid bare brackets. Now the full payload.

62. user (2026-06-22T02:02:55.101Z)

Output does not match required schema: root: must have required property 'next_items', root: must have required property 'stale_doc_corrections'

63. assistant (2026-06-22T02:03:00.686Z)

Failed again ? so there are MORE `[]`/bracket-like tokens or another breaker in the full payload. The `why` fields contain `(:514-516)` and `tools/cleanup\_caches.py` etc. ? but more likely there's still a bare `[` somewhere, OR the `current\_status` still has a subtle issue. The last successful call had a simple `current\_status`. Let me bisect the full payload: keep the FULL `next\_items` + `stale\_doc\_corrections` but use a minimal `current\_status`, to see if the arrays are the problem.

64. user (2026-06-22T02:03:14.272Z)

```
<tool_use_error>InputValidationError: StructuredOutput was called with input that could not be
parsed as JSON.
You sent (first 200 of 4291 bytes): {"area": "This-session follow-ups + cross-cutting hygiene",
"current_status": "See next_items and stale_doc_corrections for the grounded detail.",
"next_items": [{"title": "Finding B: fail the job (or
Common causes: unescaped backslashes in file paths (use / or \\), unescaped control characters,
or truncated output. Retry with valid JSON.</tool_use_error>
```

65. assistant (2026-06-22T02:03:17.950Z)

That `\_\_unparsedToolInput` path isn't valid. Let me go back to the proper parameter form and
bisect: full arrays with a trivial `current\_status` (no brackets).

66. user (2026-06-22T02:03:32.914Z)

```
<tool_use_error>InputValidationError: StructuredOutput was called with input that could not be
parsed as JSON.
You sent (first 200 of 4292 bytes): {"area": "This-session follow-ups + cross-cutting hygiene",
"current_status": "See next_items and stale_doc_corrections for the grounded detail.",
"next_items": [{"title": "Finding B: fail the job on
Common causes: unescaped backslashes in file paths (use / or \\), unescaped control characters,
or truncated output. Retry with valid JSON.</tool_use_error>
```

67. assistant (2026-06-22T02:03:36.773Z)

Let me stop using `\_\_unparsedToolInput` and go back to the named parameters that worked. I'll
bisect with the real parameters: minimal current_status (no brackets) + the FULL next_items
array + full corrections, to isolate whether a bracket lurks in the arrays.

68. user (2026-06-22T02:03:52.849Z)

Structured output provided successfully

69. assistant (2026-06-22T02:04:01.678Z)

The full payload succeeded. The breaker throughout was the bare `[]` empty-list literal inside
the `current\_status` string (the harness parsed `[` as the start of an array). With it spelled
out as "empty list", everything validated.

Final grounded findings have been returned via StructuredOutput. Summary of what I surfaced for
the co-founder:

****P0 ? Finding B (real, untracked):**** A missing `GEMINI\_API\_KEY` (or any unrunnable config)
lands as a 0-rally `status=done` "success" ? verified at `gemini.py:92-99` + `:514-516` and
`job\_worker.py:76-90`. The error only hits the log. Should fail the job (or surface a structured
error the SPA shows). Not in BACKLOG/DOC_STATUS/NEXT_STEPS; needs a tracked one-PR hardening
item.

****P1 follow-ups:****

- Auto-schedule the retention sweep on the droplet (engine #301 open; #299 shipped only the dry-run tool).
- Revoke the broad account-wide cloudflared `cert.pem` left on the box (DEPLOY_ALPHA \$5/\$8) ? owner action.
- Native-speaker review of es/da/id + watermark strings + owner kn/hi font eyeball (wide-launch gate) ? owner-coordinated.

****P2:**** Basic live-droplet monitoring (the Ops-Agent #291 is GPU-VM-only, not this CPU droplet).

I also corrected five stale/half-true doc claims (the #294 preflight does NOT cover Finding B;
retention is only half-done; Ops-Agent doesn't cover the droplet; #301 is still open).